MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, ADVANCED LOW-POWER SCHOTTKY TTL, BUS TRANSCEIVERS WITH THREE-STATE OUTPUTS, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, advanced low-power Schottky TTL, bus transceivers with three-state outputs. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 Device types. The device types shall be as follows:

Device type	Circuit
01	Octal bus transceivers with three-state outputs
02	Octal bus transceivers with open-collector outputs
03	Octal bus transceivers with open-collector outputs
04	Octal bus transceivers with three-state outputs
05	Octal bus transceivers with three-state outputs
06	Quadruple bus transceivers with three-state outputs
07	Quadruple bus transceivers with three-state outputs

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
C	D-1 (14-lead, 1/4" x 3/4"), dual-in-line package
D	F-2 (14-lead, 1/4" x 3/8"), flat-package
R	D-8 (20-lead, 1/4" x 1 1/16"), dual-in-line package
S	F-9 (20-lead, 1/4" x 1/2"), flat package
X	C-2A (20-terminal, .350" x .350"), square chip carrier package
2	C-2 (20-terminal, .350" x .350"), square chip carrier package

1.3 Absolute maximum ratings.

Supply voltage range (V_{CC})	-0.5 V dc to +7.0 V dc
Input voltage range	-1.5 V dc at -18 mA to +5.5 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation per device (P_D): 1/	
Device type 01	264 mA
Device type 02	247.5 mA
Device type 03	154 mA
Device type 04	291.5 mA
Device type 05	346.5 mA
Device types 06 and 07	176 mA

1/ Must withstand the added P_D due to short circuit test (e.g., I_O).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441-5700, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC}):	
Cases C, D, R, S, X, and 2	(See MIL-M-38510, appendix C)
Junction temperature (T_J) 2/	+175°C

1.4 Recommended operating conditions.

Supply voltage (V_{CC})	4.5 V dc minimum to 5.5 V dc maximum
Minimum high level input voltage (V_{IH})	2.0 V dc
Maximum low level input voltage (V_{IL})	0.8 V dc
Normalized fanout (each output) 3/	20 maximum
Case operating temperature range (T_C)	-55°C to +125°C

2 APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specification and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510, and specified herein.

3.2.1 Terminal connections and logic diagrams. The terminal connections and logic diagrams shall be as specified on figures 1 and 2, respectively.

3.2.2 Truth table. The truth table shall be as specified on figure 3.

3.2.3 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request.

2/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions in accordance with method 5004 of MIL-STD-883.

3/ The device shall fanout in both high and low levels to the specified number of inputs of the same device type as that being tested.

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3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.4).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C < T_C < +125°C$ unless otherwise specified		Device type	Limits		Unit
					Min	Max	
High level output voltage	V_{OH1}	V_{CC} = 4.5 V V_{IN} = 0.8 V or 2.0 V I_{OH} = -3 mA		01,04 05,06 07	2.4		V
	V_{OH2}	V_{CC} = 4.5 V V_{IN} = 0.8 V or 2.0 V I_{OH} = -12 mA		01,04 05,06 07	2.0		V
High level output current	I_{OH}	V_{CC} = 4.5 V V_{OH} = 5.5 V		02,03		100	µA
Low level output voltage	V_{OL}	V_{CC} = 4.5 V V_{IN} = 0.8 V or 2.0 V I_{OL} = 12 mA		All		0.4	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, T_C = +25°C I_I = -18 mA		All		-1.5	V
Low level input current	I_{IL}	V_{CC} = 5.5 V V_{IN} = 0.4 V	DIR and $\overline{G}$ $\overline{G}$AB and GBA	All	0	-100	µA
High level input current	I_{IH1}	V_{CC} = 5.5 V V_{IN} = 2.7 V	DIR and $\overline{G}$ $\overline{G}$AB and GBA	All		20	µA
	I_{IH2}	V_{CC} = 5.5 V V_{IN} = 5.5 V	I/O ports	All		100	µA
	I_{IH3}	V_{CC} = 5.5 V V_{IN} = 7.0 V	DIR and $\overline{G}$ $\overline{G}$AB and GBA	All		100	µA
Output current, high level outputs off	I_{OZH}/ I_{IH1}	V_{CC} = 5.5 V V_O = 2.7 V	I/O ports	01,04 05		20	µA
Output current, low level outputs off	I_{OZL}/ I_{IL}	V_{CC} = 5.5 V V_O = 0.4 V	I/O ports	01,04 05		-100	µA
Output current 1/	I_O	V_{CC} = 5.5 V V_O = 2.25 V		01,04 05	-30	-112	mA

See footnote at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C < T_C < +125°C$ unless otherwise specified		Device type	Limits Min	Limits Max	Unit
Supply current	I_{CCH}	V_{CC} = 5.5 V		01		36	mA
				02		37	
				03		15	
				04		41	
				05		48	
				06		20	
				07		25	
Supply current outputs low	I_{CCL}	V_{CC} = 5.5 V		01,02		45	mA
				03		28	
				04		51	
				05		60	
				06		26	
				07		30	
Supply current outputs disabled	I_{CCZ}	V_{CC} = 5.5 V		01		48	mA
				04		53	
				05		63	
				06		27	
				07		32	
Propagation delay time low-to-high level A or B to B or A	t_{PLH1}	V_{CC} = 5.0 V, C_L = 50 pF	$R_1 = R_2$ = 500Ω	01,05	2	14	ns
				04,06	3	15	
				07	4	15	
			R_L = 680Ω	02	10	37	
				03	15	50	
Propagation delay time high-to-low level A or B to B or A	t_{PHL1}	V_{CC} = 5.0 V, C_L = 50 pF	$R_1 = R_2$ = 500Ω	01,04	2	13	ns
				05	3	13	
				06	2	14	
				07	4	15	
			R_L = 680Ω	02	5	23	
				03	5	25	
Propagation delay time low-to-high level $\overline{G}$ or DIR to A or B	t_{PLH2}	V_{CC} = 5.0 V, C_L = 50 pF, R_L = 680Ω		02	13	40	ns
				03	15	45	
Propagation delay time high-to-low level $\overline{G}$ or DIR to A or B	t_{PHL2}			02	13	40	
				03	15	50	
Propagation delay time disable to high level $\overline{G}$ to A or B	t_{PZH}	V_{CC} = 5.0 V, C_L = 50 pF, $R_1 = R_2$ = 500Ω		01	5	25	ns
				04	6	28	
				05,07	3	25	
				06	4	22	
Propagation delay time disable to low level $\overline{G}$ to A or B	t_{PZL}			01,05	4	27	
				04	8	28	
				06,07	7	25	
Propagation delay time high level to disable G to A or B	t_{PHZ}	V_{CC} = 5.0 V, C_L = 50 pF, $R_1 = R_2$ = 500Ω		01	3	15	ns
				04	2	17	
				05	3	16	
				06,07	2	16	
Propagation delay time low level to disable G to A or B	t_{PLZ}			01,05	4	18	
				04	4	33	
				06	4	28	
				07	4	30	

1/ The output conditions have been chosen to produce a current that closely approximates one half of the true short-circuit output current, I_{OS}.

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3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 9 (see MIL-M-38510, appendix E).

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,7,9,10,11	1*,2,3,7,9
Group A test requirements (method 5005)	1,2,3,7,8,9,10,11	1,2,3,7,8,9,10,11
Group B test requirements (method 5005) subgroup 5	1,2,3,7,8,9,10,11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,2,3
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3

* PDA applies to subgroup 1 (see 4.2c).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in (method 1015 of MIL-STD-883).

(1) Test condition D, using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

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4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Electrical test requirements shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical test requirements shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D, using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the reference terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

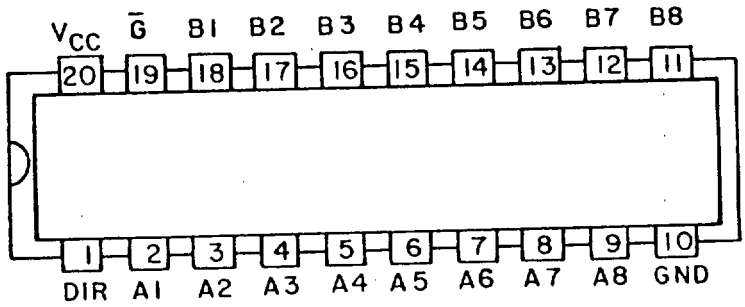
f. Requirements for product assurance options.

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Device types 01, 02, 03, 04, and 05

Cases R and S



Device types 01, 02, 03, 04, and 05

Cases 2 and X

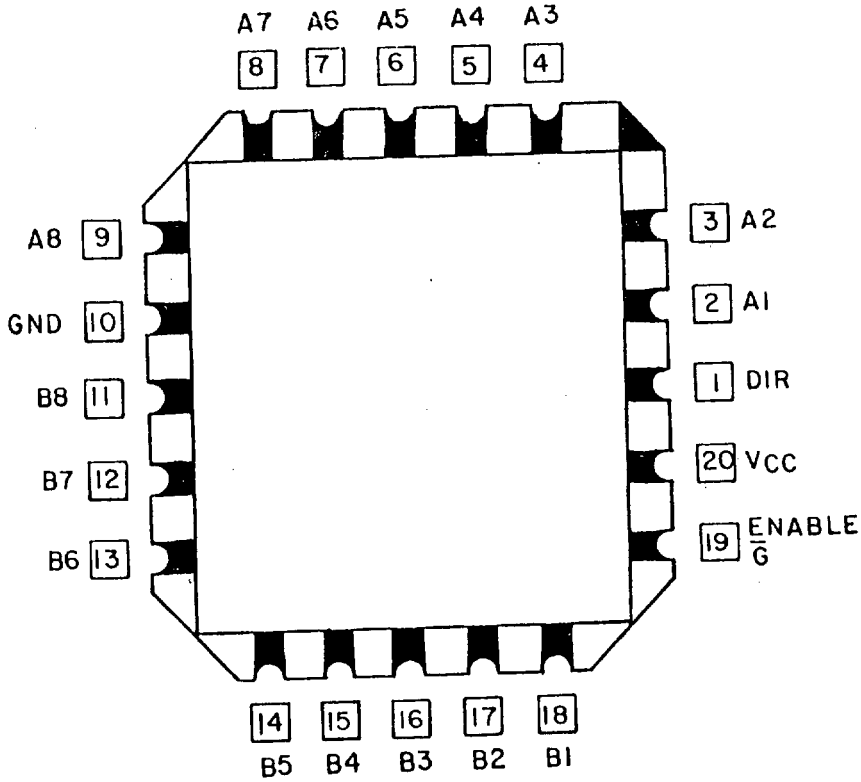
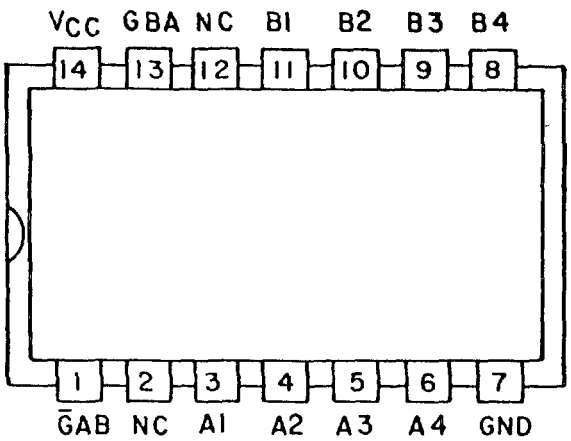


FIGURE 1. Terminal connections (top view).

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Device types 06 and 07

Cases C and D



Device types 06 and 07

Cases 2 and X

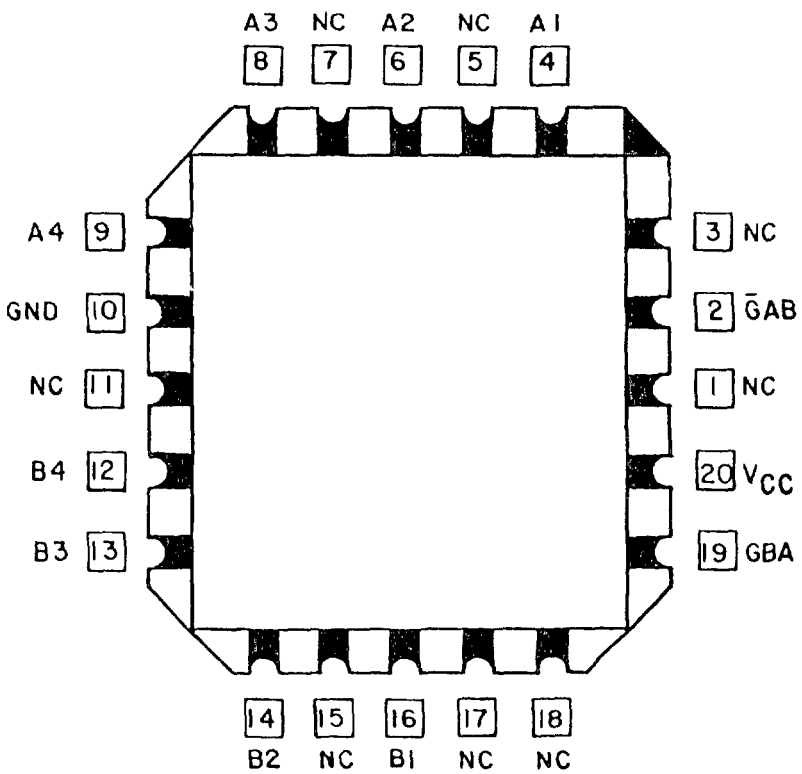
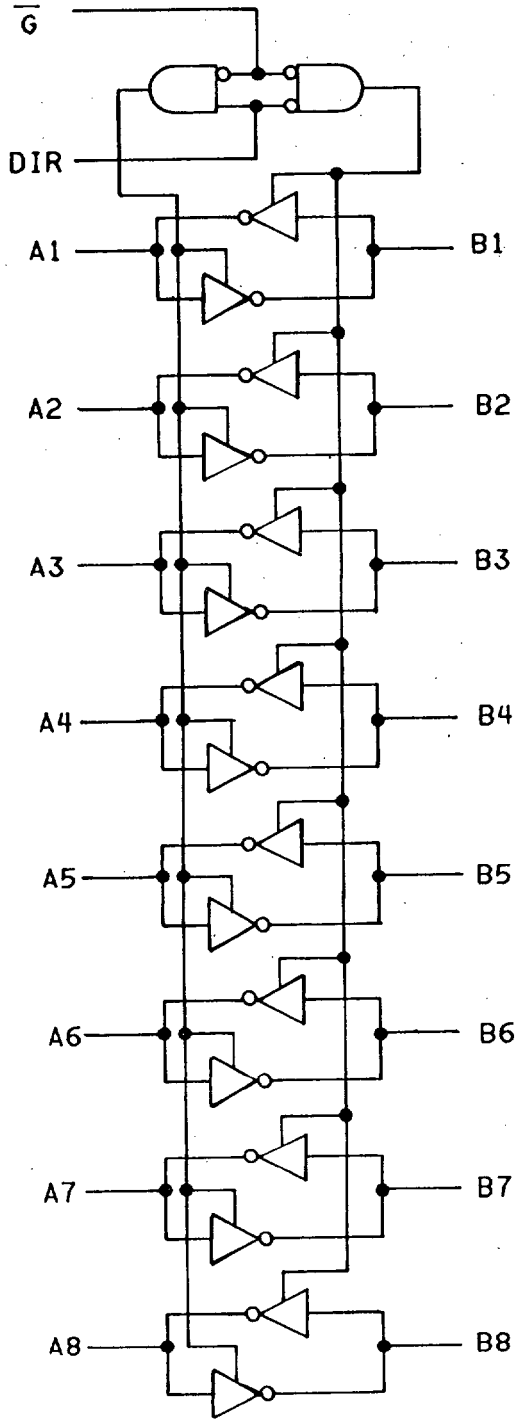


FIGURE 1. Terminal connections (top view) - Continued.

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Device type 01



Device type 02

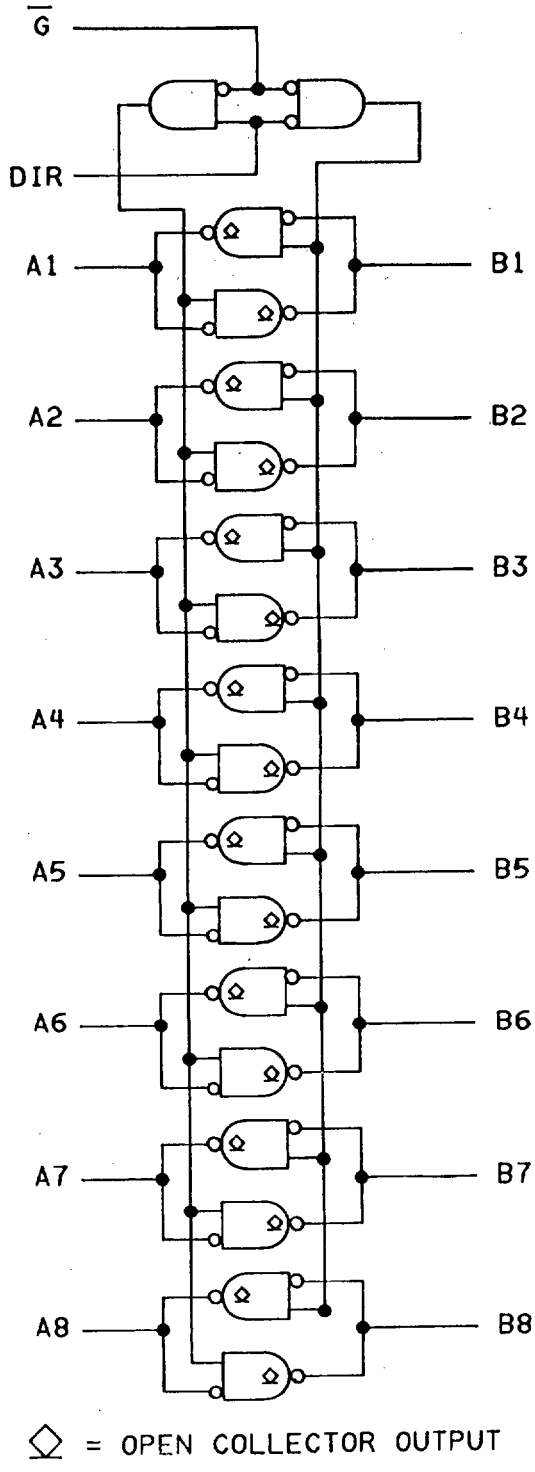
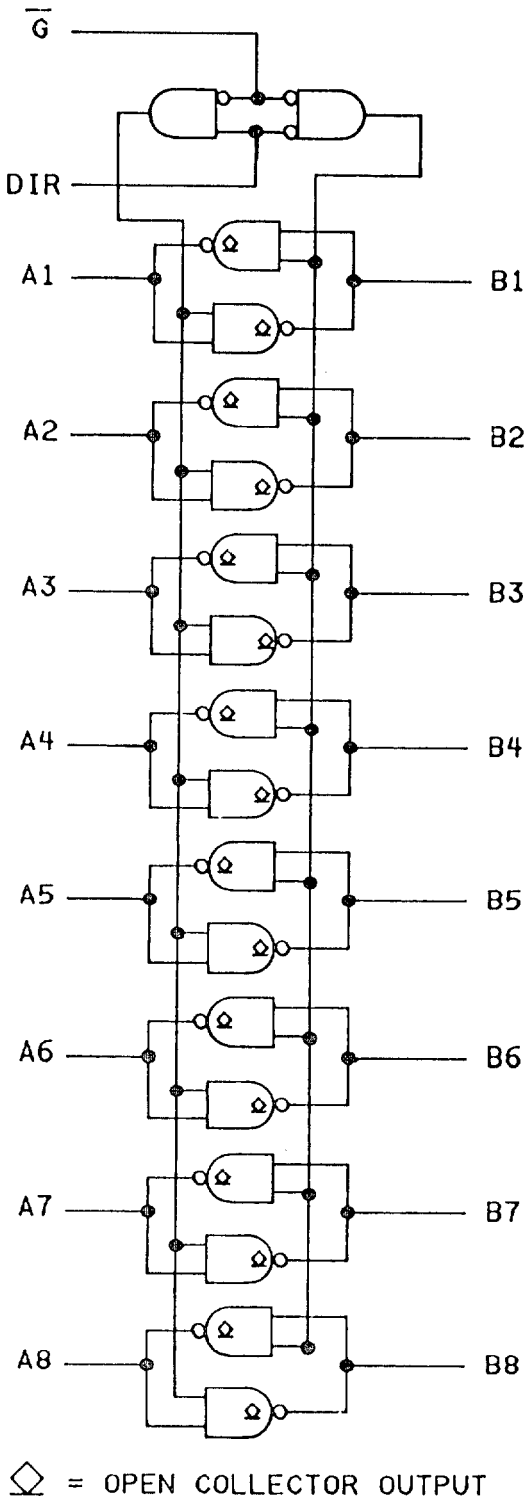


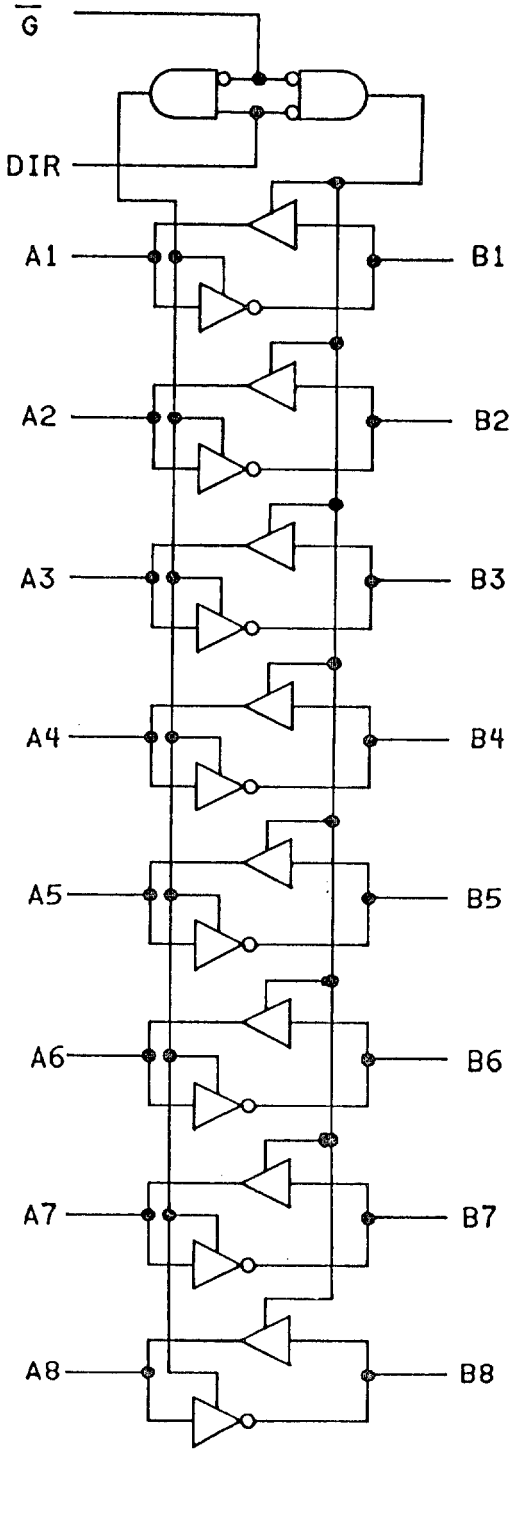
FIGURE 2. Logic diagrams.

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Device type 03



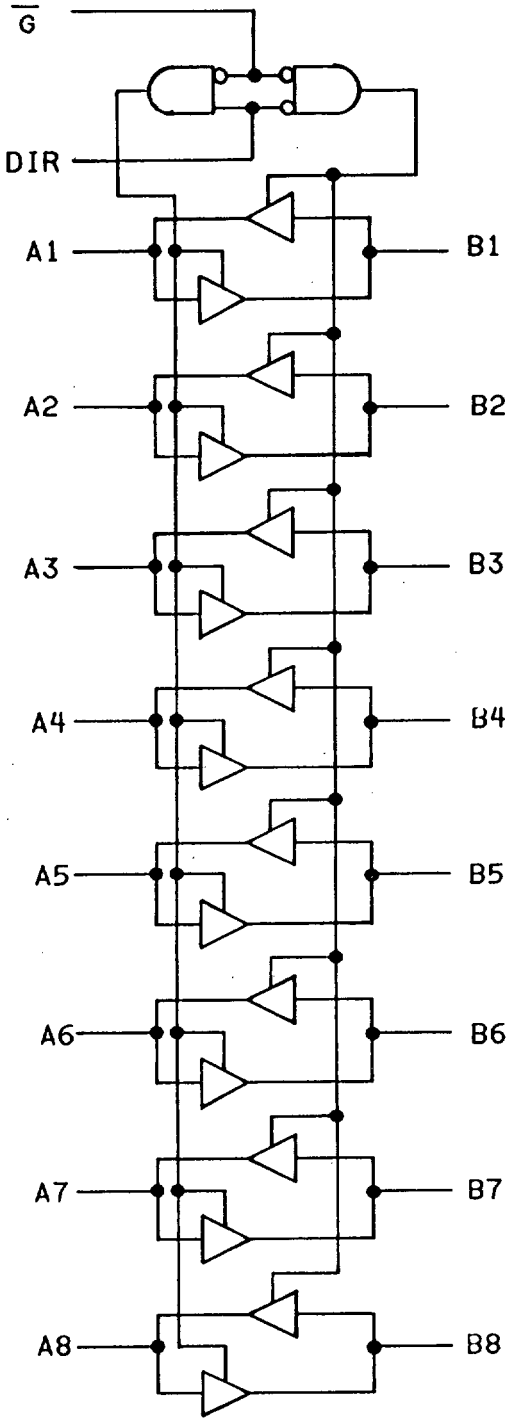
Device type 04



◇ = OPEN COLLECTOR OUTPUT

FIGURE 2. Logic diagrams - Continued.

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Device type 06

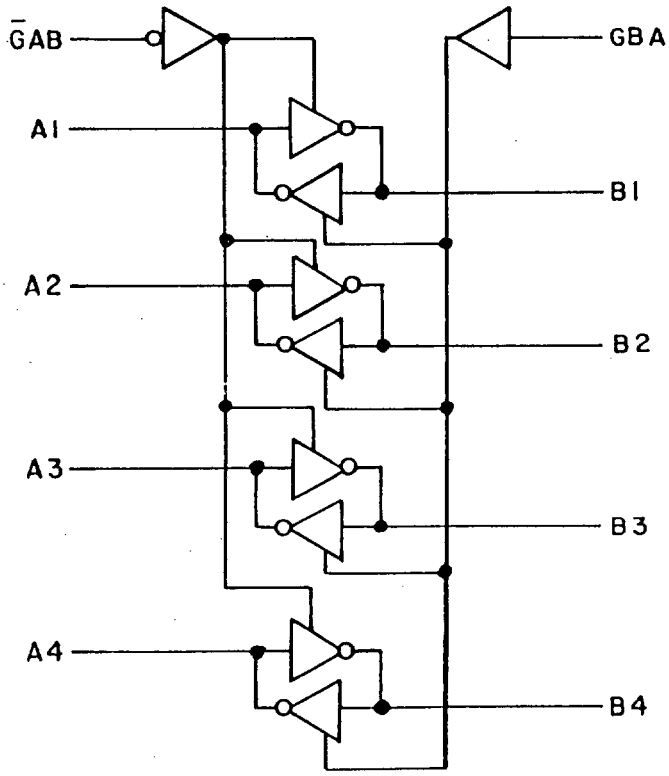


FIGURE 2. Logic diagrams - Continued.

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Device type 07

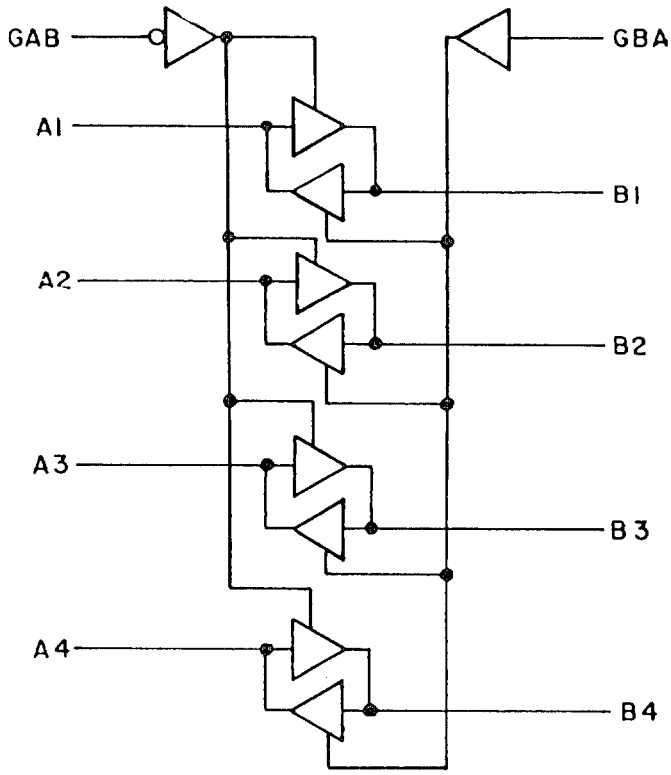


FIGURE 2. Logic diagrams - Continued.

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Device types 01 and 03

Enable $\overline{G}$	Direction Control Dir	Operation
L	L	$\overline{B}$ data to A bus
L	H	$\overline{A}$ data to B bus
H	X	Isolation

H = High level, L = Low level, X = Irrelevant

Device types 02 and 05

Enable $\overline{G}$	Direction Control Dir	Operation
L	L	B data to A bus
L	H	A data to B bus
H	X	Isolation

H = High level, L = Low level, X = Irrelevant

Device type 04

Enable $\overline{G}$	Direction Control Dir	Operation
L	L	B data to A bus
L	H	$\overline{A}$ data to B bus
H	X	Isolation

H = High level, L = Low level, X = Irrelevant

FIGURE 3. Truth tables.

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Device type 06

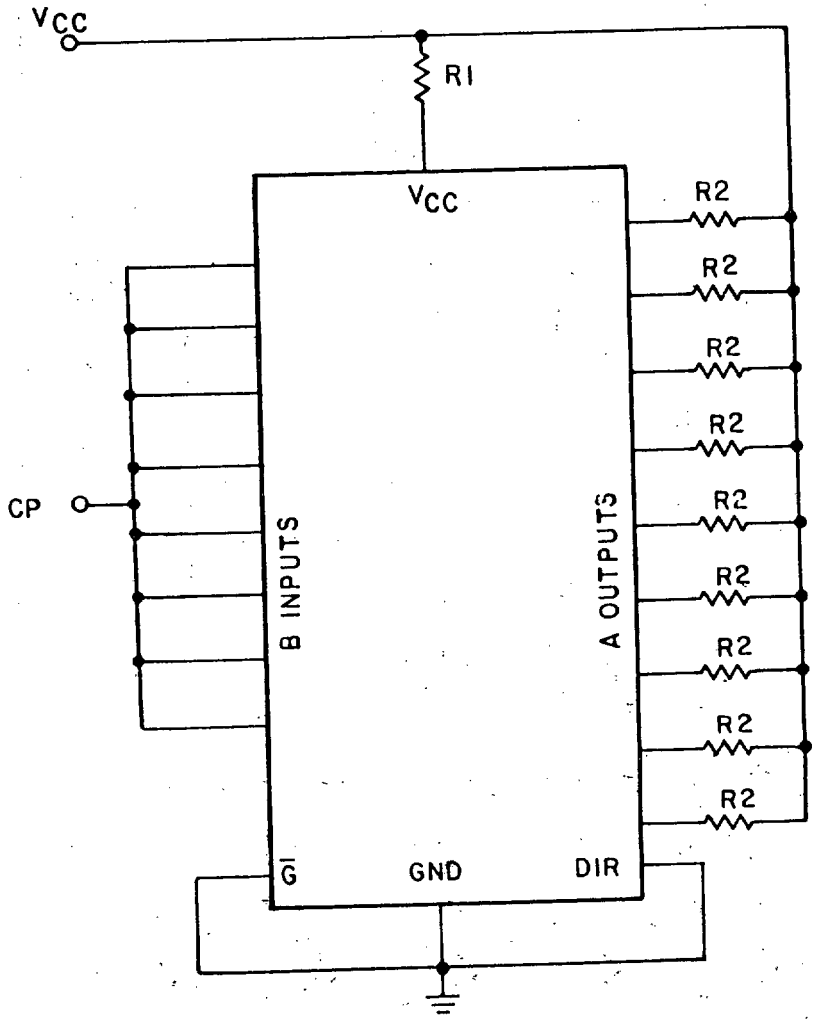
Inputs		Outputs
$\overline{G}AB$	GBA	
L	L	$\overline{A}$ to B
H	H	$\overline{B}$ to A
H	L	Isolation
L	H	Latch A and B ($A = \overline{B}$)

Device type 07

Inputs		Outputs
$\overline{G}AB$	GBA	
L	L	A to B
H	H	B to A
H	L	Isolation
L	H	Latch A and B (A = B)

FIGURE 3. Truth tables - Continued.

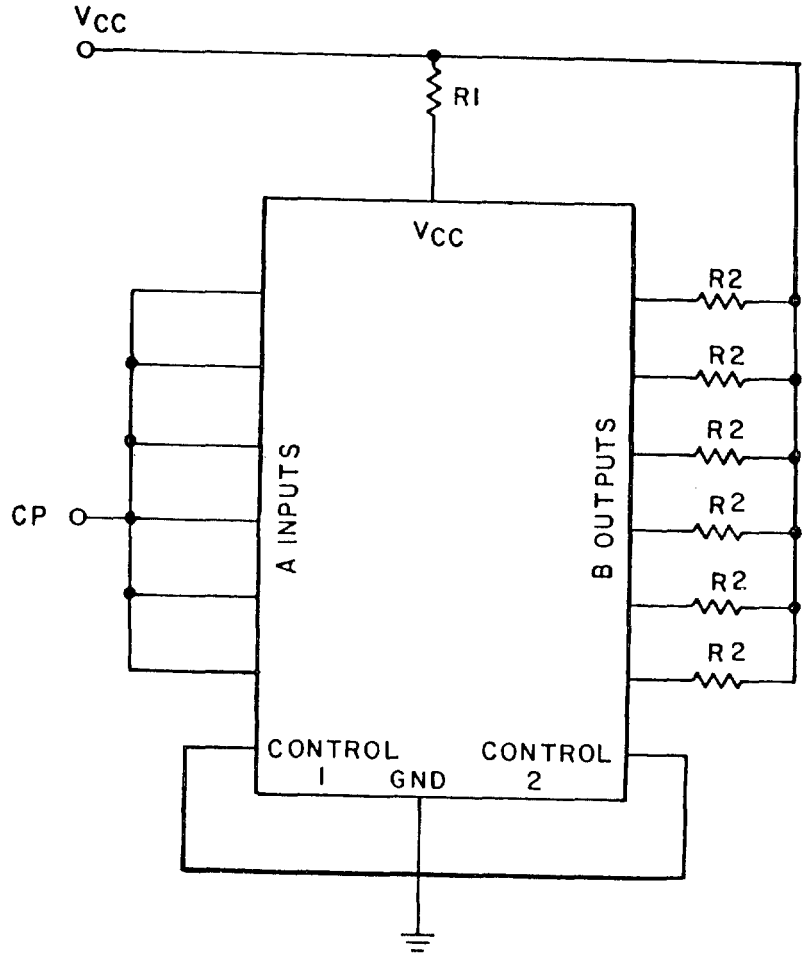
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NOTES:
1. V_{CC} and R1 shall be chosen to insure a 5 V minimum at device V_{CC} terminal.
2. R2 = 470Ω ±5%.
3. CP = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V to 0.8 V; V_{IH} = 2 V to 5.5 V.

FIGURE 4. Burn-in and life test circuit.

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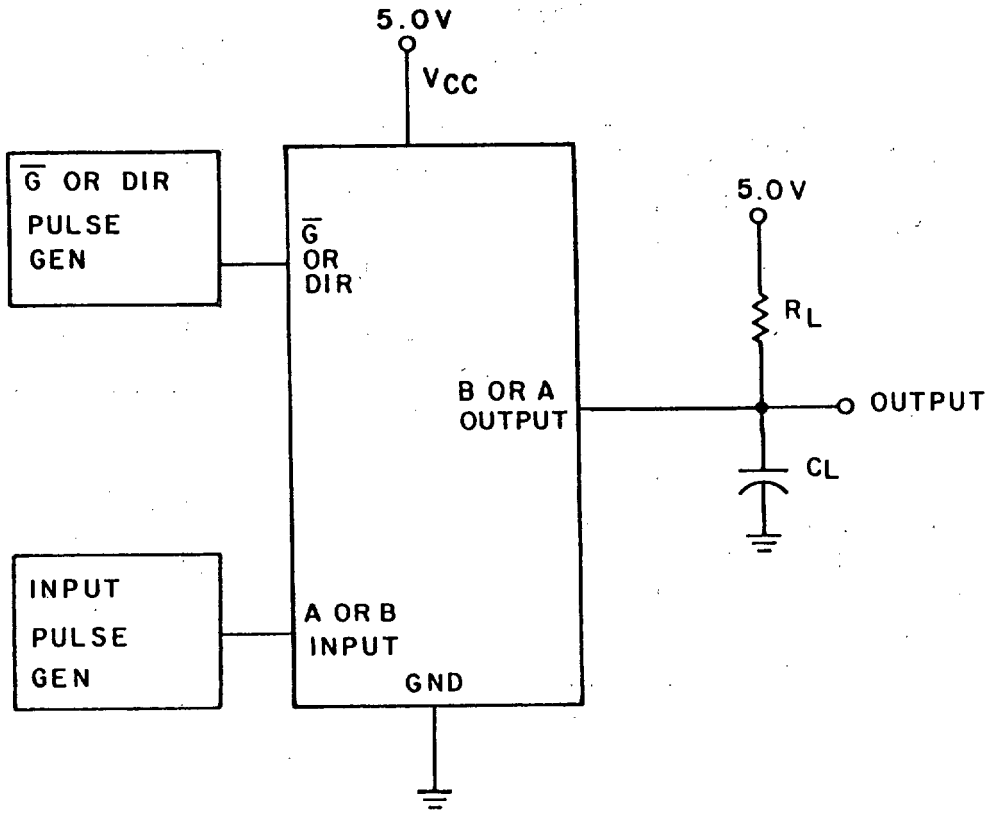
NOTES:

1. V_{CC} and RI shall be chosen to insure a 5 V minimum at device V_{CC} terminal.
2. R2 = 470Ω ±5%.
3. CP = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V to 0.8 V; V_{IH} = 2 V to 5.5 V.

FIGURE 4. Burn-in and life test circuit - Continued.

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Device types 02 and 03



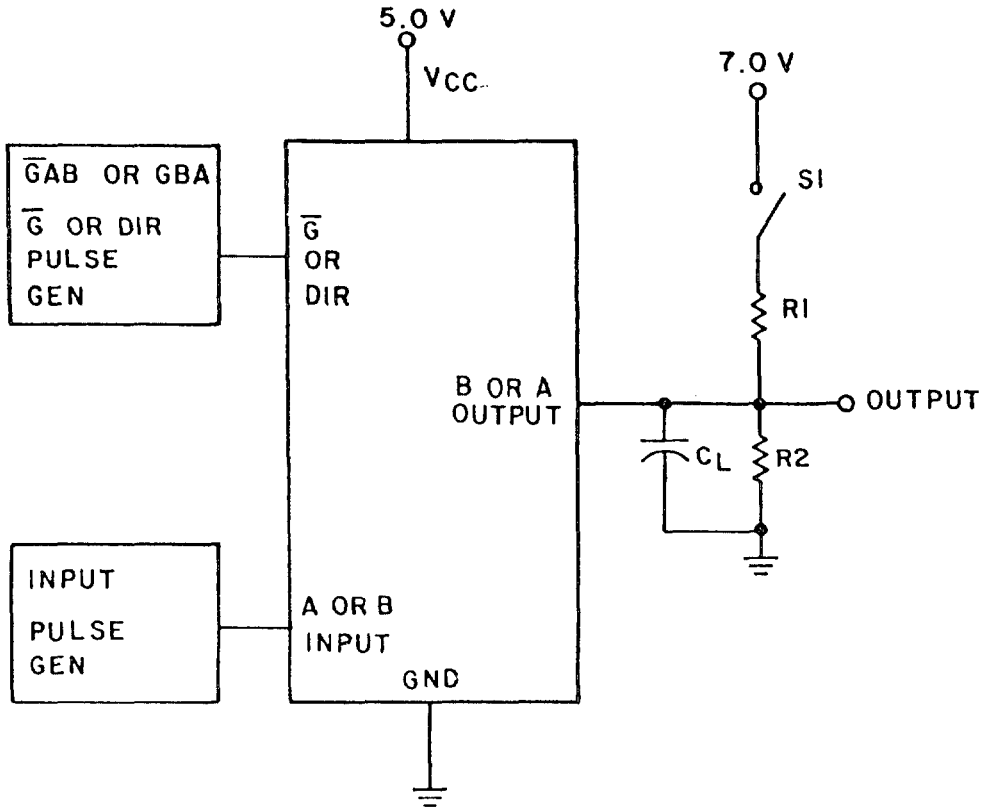
TEST CIRCUIT

NOTES:
1. The input pulse shall have the following characteristics: $t_1 = t_0 = 6$ ns ±1.5 ns; PRR ≤ 1 MHz.
2. $R_L = 680\Omega$ ±5%.
3. $C_L = 50$ pF ±10%, including scope probe, wiring, and stray capacitance without package in test fixture.
4. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 5. Switching time test circuit - Continued.

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Device types 01, 04, 05, 06, and 07



TEST CIRCUIT

NOTES:
1. The input pulse shall have the following characteristics:
   $t_1 = t_0$ = 6 ns ±1.5 ns; PRR ≤ 1 MHz.
2. R1 = R2 = 499Ω ±1%.
3. C_L = 50 pF ±10%, including scope probe, wiring, and stray capacitance without package in test fixture.
4. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 5. Switching time test circuit.

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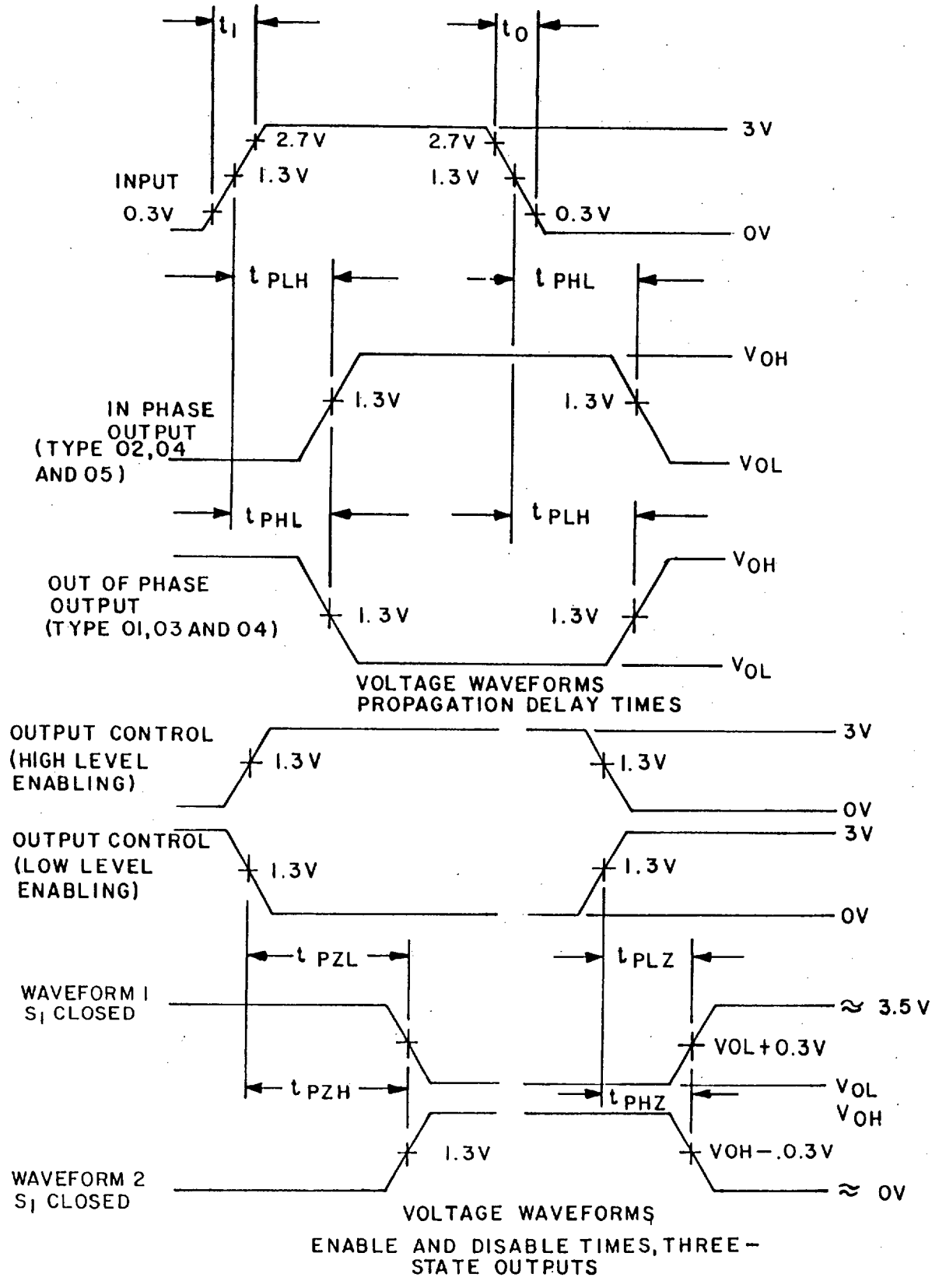


FIGURE 5. Switching time test circuit - Continued.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,X,Z Test no.	1 DIR	2 A1	3 A2	4 A3	5 A4	6 A5	7 A6	8 A7	9 A8	10 GND	11 B8	12 B7	13 B6	14 B5	15 B4	16 B3	17 B2	18 B1	19 $\overline{G}$	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	V_{OH1}	3006	1	2.0 V	0.8 V	0.8 V							GND							-3 mA	-3 mA	0.8 V	4.5 V	B1	2.4		V
			2	"	"		0.8 V						"						-3 mA	"	"	"	"	B2	"		"
			3	"	"			0.8 V					"				-3 mA		"	"	"	"	"	B3	"		"
			4	"	"				0.8 V				"			-3 mA	"		"	"	"	"	"	B4	"		"
			5	"	"					0.8 V			"		-3 mA	"	"		"	"	"	"	"	B5	"		"
			6	"	"						0.8 V		"	-3 mA	"	"	"		"	"	"	"	"	B6	"		"
			7	"	"							0.8 V	"	"	"	"	"		"	"	"	"	"	B7	"		"
			8	"	"								"						"	"	"	"	"	B8	"		"
			9	0.8 V	-3 mA								"						"	0.8 V	"	"	"	A1	"		"
			10	"	"	-3 mA							"						0.8 V	"	"	"	"	A2	"		"
			11	"	"		-3 mA						"						"	"	"	"	"	A3	"		"
			12	"	"			-3 mA					"				0.8 V		"	"	"	"	"	A4	"		"
			13	"	"				-3 mA				"			0.8 V	"		"	"	"	"	"	A5	"		"
			14	"	"					-3 mA			"		0.8 V	"	"		"	"	"	"	"	A6	"		"
			15	"	"						-3 mA		"	0.8 V	"	"	"		"	"	"	"	"	A7	"		"
			16	"	"							-3 mA	"	"	"	"	"		"	"	"	"	"	A8	"		"
	V_{OH2}	3006	17	2.0 V	0.8 V	0.8 V							"						-12 mA	-12 mA	"	"	"	B1	2.0		"
			18	"	"		0.8 V						"						"	"	"	"	"	B2	"		"
			19	"	"			0.8 V					"				-12 mA		"	"	"	"	"	B3	"		"
			20	"	"				0.8 V				"			-12 mA	"		"	"	"	"	"	B4	"		"
			21	"	"					0.8 V			"		-12 mA	"	"		"	"	"	"	"	B5	"		"
			22	"	"						0.8 V		"	-12 mA	"	"	"		"	"	"	"	"	B6	"		"
			23	"	"							0.8 V	"	"	"	"	"		"	"	"	"	"	B7	"		"
			24	"	"								"						"	"	"	"	"	B8	"		"
			25	0.8 V	-12 mA	-12 mA							"						"	0.8 V	"	"	"	A1	"		"
			26	"	"		-12 mA						"						0.8 V	"	"	"	"	A2	"		"
			27	"	"			-12 mA					"				0.8 V		"	"	"	"	"	A3	"		"
			28	"	"				-12 mA				"			0.8 V	"		"	"	"	"	"	A4	"		"
			29	"	"					-12 mA			"		0.8 V	"	"		"	"	"	"	"	A5	"		"
			30	"	"						-12 mA		"	0.8 V	"	"	"		"	"	"	"	"	A6	"		"
			31	"	"							-12 mA	"	"	"	"	"		"	"	"	"	"	A7	"		"
			32	"	"								"						"	"	"	"	"	A8	"		"
	V_{OL}	3007	33	2.0 V	2.0 V	2.0 V							"						12 mA	12 mA	"	"	"	B1		0.4	"
			34	"	"		2.0 V						"						"	"	"	"	"	B2		"	"
			35	"	"			2.0 V					"				12 mA		"	"	"	"	"	B3		"	"
			36	"	"				2.0 V				"			12 mA	"		"	"	"	"	"	B4		"	"
			37	"	"					2.0 V			"		12 mA	"	"		"	"	"	"	"	B5		"	"
			38	"	"						2.0 V		"	12 mA	"	"	"		"	"	"	"	"	B6		"	"
			39	"	"							2.0 V	"	"	"	"	"		"	"	"	"	"	B7		"	"
			40	"	"								"						"	"	"	"	"	B8		"	"
			41	0.8 V	12 mA	12 mA							"						2.0 V	2.0 V	"	"	"	A1		"	"
			42	"	"		12 mA						"						"	"	"	"	"	A2		"	"
			43	"	"			12 mA					"				2.0 V		"	"	"	"	"	A3		"	"
			44	"	"				12 mA				"			2.0 V	"		"	"	"	"	"	A4		"	"
			45	"	"					12 mA			"		2.0 V	"	"		"	"	"	"	"	A5		"	"
			46	"	"						12 mA		"	2.0 V	"	"	"		"	"	"	"	"	A6		"	"
			47	"	"							12 mA	"	"	"	"	"		"	"	"	"	"	A7		"	"
			48	"	"								"						"	"	"	"	"	A8		"	"
	V_{IC}		49	-18 mA									"											DIR		-1.5	"
			50	"	-18 mA								"											A1		"	"
			51	"	"	-18 mA							"											A2		"	"
			52	"	"		-18 mA						"											A3		"	"
			53	"	"			-18 mA					"											A4		"	"
			54	"	"				-18 mA				"											A5		"	"
			55	"	"					-18 mA			"											A6		"	"
			56	"	"						-18 mA		"											A7		"	"
			57	"	"							-18 mA	"											A8		"	"
			58	"	"								"	-18 mA										B8		"	"
			59	"	"								"		-18 mA									B7		"	"
			60	"	"								"			-18 mA								B6		"	"
			61	"	"								"				-18 mA							B5		"	"
			62	"	"								"					-18 mA						B4		"	"
			63	"	"								"						-18 mA					B3		"	"
			64	"	"								"							-18 mA				B2		"	"
			65	"	"								"								-18 mA			B1		"	"
			66	"	"								"											$\overline{G}$		"	"

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases (R,S,X,Z) Test no.	1 DIR	2 A1	3 A2	4 A3	5 A4	6 A5	7 A6	8 A7	9 A8	10 GND	11 B8	12 B7	13 B6	14 B5	15 B4	16 B3	17 B2	18 B1	19 $\overline{G}$	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
T_C = +25°C 1/	I_{IL}	3009	67	0.4 V									GND									0.4 V	5.5 V	DIR	0	-100	µA
	I_{IL}	3009	68										"									"	"	$\overline{G}$	"	"	"
	I_{IL}/ I_{OZL}		69		0.4 V								"									4.5 V	"	A1	"	"	"
			70			0.4 V							"									"	"	A2	"	"	"
			71				0.4 V						"									"	"	A3	"	"	"
			72					0.4 V					"									"	"	A4	"	"	"
			73						0.4 V				"									"	"	A5	"	"	"
			74							0.4 V			"									"	"	A6	"	"	"
			75								0.4 V		"									"	"	A7	"	"	"
			76									0.4 V	"									"	"	A8	"	"	"
			77										"	0.4 V								"	"	B8	"	"	"
			78										"		0.4 V							"	"	B7	"	"	"
			79										"			0.4 V						"	"	B6	"	"	"
			80										"				0.4 V					"	"	B5	"	"	"
			81										"					0.4 V				"	"	B4	"	"	"
			82										"						0.4 V			"	"	B3	"	"	"
			83										"							0.4 V		"	"	B2	"	"	"
			84										"									"	"	B1	"	"	"
	I_{IH1}	3010	85	2.7 V									"									2.7 V	"	DIR		20	"
	I_{IH1}	3010	86										"									"	"	$\overline{G}$		"	"
	I_{IH1}/ I_{OZH}		87		2.7 V								"									4.5 V	"	A1		"	"
			88			2.7 V							"									"	"	A2		"	"
			89				2.7 V						"									"	"	A3		"	"
			90					2.7 V					"									"	"	A4		"	"
			91						2.7 V				"									"	"	A5		"	"
			92							2.7 V			"									"	"	A6		"	"
			93								2.7 V		"									"	"	A7		"	"
			94									2.7 V	"									"	"	A8		"	"
			95										"	2.7 V								"	"	B8		"	"
			96										"		2.7 V							"	"	B7		"	"
			97										"			2.7 V						"	"	B6		"	"
			98										"				2.7 V					"	"	B5		"	"
			99										"					2.7 V				"	"	B4		"	"
			100										"						2.7 V			"	"	B3		"	"
			101										"							2.7 V		"	"	B2		"	"
			102										"									"	"	B1		"	"
	I_{IH2}	3010	103		5.5 V								"										"	A1		100	"
			104			5.5 V							"										"	A2		"	"
			105				5.5 V						"										"	A3		"	"
			106					5.5 V					"										"	A4		"	"
			107						5.5 V				"										"	A5		"	"
			108							5.5 V			"										"	A6		"	"
			109								5.5 V		"										"	A7		"	"
			110									5.5 V	"										"	A8		"	"
			111										"	5.5 V									"	B8		"	"
			112										"		5.5 V								"	B7		"	"
			113										"			5.5 V							"	B6		"	"
			114										"				5.5 V						"	B5		"	"
			115										"					5.5 V					"	B4		"	"
			116										"						5.5 V				"	B3		"	"
			117										"							5.5 V			"	B2		"	"
			118										"										"	B1		"	"
	I_{IH3}	3010	119	7.0 V									"									7.0 V	"	DIR		"	"
		3010	120										"									"	"	$\overline{G}$		"	"
	I_{O} 1/	3011	121	4.5 V	GND								"									GND	"	B1	2/ -30	2/ -112	mA
			122	"		GND							"									"	"	B2	"	"	"
			123	"			GND						"									"	"	B3	"	"	"
			124	"				GND					"									"	"	B4	"	"	"
			125	"					GND				"									"	"	B5	"	"	"
			126	"						GND			"									"	"	B6	"	"	"
			127	"							GND		"									"	"	B7	"	"	"
			128	"								GND	"	2.25 V								"	"	B8	"	"	"
															2.25 V												
																2.25 V											
																	2.25 V										
																		2.25 V									
																			2.25 V								
																				2.25 V							

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,X,2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				DIR	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	G	V_{CC}				
1 T_C = +25°C	I_O 1/	3011	129	GND	2.25 V								GND											A1	2/ -30	2/ -112	mA
			130	"		2.25 V							"											A2	"	"	"
			131	"			2.25 V						"											A3	"	"	"
			132	"				2.25 V					"											A4	"	"	"
			133	"					2.25 V				"											A5	"	"	"
			134	"						2.25 V			"											A6	"	"	"
			135	"							2.25 V		"											A7	"	"	"
			136	"								2.25 V	"	GND										A8	"	"	"
	I_{CCH}	3005	137	5.5 V	GND	GND	GND	GND	GND	GND	GND	GND	"									"	"	V_{CC}		36	"
	I_{CCL}	3005	138	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"										"	"	V_{CC}		45	"
	I_{CCZ}	3005	139	5.5 V									"									5.5 V	"	V_{CC}		66	"
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C and omit V_{IC} tests.																										
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and omit V_{IC} tests.																										
7 T_C = +25°C	Truth table tests	3014	140	A 3/	A	A	A	A	A	A	A	A	GND	L	L	L	L	L	L	L	L	B	4.5 V	See 4/			
			141	A	B 3/	B	B	B	B	B	B	B	"	H	H	H	H	H	H	H	H	"	"				
			142	B	L	L	L	L	L	L	L	L	"	A	A	A	A	A	A	A	A	"	"				
			143	B	H	H	H	H	H	H	H	H	"	B	B	B	B	B	B	B	B	"	"				
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																										
9 T_C = +25°C	t_{PLH1}	3003 (fig. 5)	144	4.5 V	IN	IN	IN	IN	IN	IN	IN	IN	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	5.0 V	A1 to B1	2	11	ns
			145	"									"									"	"	A2 to B2	"	"	"
			146	"									"									"	"	A3 to B3	"	"	"
			147	"									"									"	"	A4 to B4	"	"	"
			148	"									"									"	"	A5 to B5	"	"	"
			149	"									"									"	"	A6 to B6	"	"	"
			150	"									"									"	"	A7 to B7	"	"	"
			151	"									"									"	"	A8 to B8	"	"	"
			152	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	IN	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	IN	"	"	B1 to A1	"	"	"
			153	"									"									"	"	B2 to A2	"	"	"
			154	"									"									"	"	B3 to A3	"	"	"
			155	"									"									"	"	B4 to A4	"	"	"
			156	"									"									"	"	B5 to A5	"	"	"
			157	"									"									"	"	B6 to A6	"	"	"
			158	"									"									"	"	B7 to A7	"	"	"
			159	"									"	IN	IN	IN	IN	IN	IN	IN	IN	"	"	B8 to A8	"	"	"
	t_{PHL1}	3003 (fig. 5)	160	4.5 V	IN	IN	IN	IN	IN	IN	IN	IN	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	"	A1 to B1	"	10	"
			161	"									"									"	"	A2 to B2	"	"	"
			162	"									"									"	"	A3 to B3	"	"	"
			163	"									"									"	"	A4 to B4	"	"	"
			164	"									"									"	"	A5 to B5	"	"	"
			165	"									"									"	"	A6 to B6	"	"	"
			166	"									"									"	"	A7 to B7	"	"	"
			167	"									"									"	"	A8 to B8	"	"	"
			168	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	IN	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	IN	"	"	B1 to A1	"	"	"
			169	"									"									"	"	B2 to A2	"	"	"
			170	"									"									"	"	B3 to A3	"	"	"
			171	"									"									"	"	B4 to A4	"	"	"
			172	"									"									"	"	B5 to A5	"	"	"
			173	"									"									"	"	B6 to A6	"	"	"
			174	"									"									"	"	B7 to A7	"	"	"
			175	"									"	IN	IN	IN	IN	IN	IN	IN	IN	"	"	B8 to A8	"	"	"
	t_{PZH}	3003 (fig. 5)	176	4.5 V	GND	GND	GND	GND	GND	GND	GND	GND	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	IN	"	$\overline{G}$ to B1	5	21	"
			177	"									"									"	"	$\overline{G}$ to B2	"	"	"
			178	"									"									"	"	$\overline{G}$ to B3	"	"	"
			179	"									"									"	"	$\overline{G}$ to B4	"	"	"
			180	"									"									"	"	$\overline{G}$ to B5	"	"	"
			181	"									"									"	"	$\overline{G}$ to B6	"	"	"
			182	"									"									"	"	$\overline{G}$ to B7	"	"	"
			183	"									"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	"	$\overline{G}$ to B8	"	"	"

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,X,Z Test no.	1 DIR	2 A1	3 A2	4 A3	5 A4	6 A5	7 A6	8 A7	9 A8	10 GND	11 B8	12 B7	13 B6	14 B5	15 B4	16 B3	17 B2	18 B1	19 $\overline{G}$	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
9 T_C = +25°C	t_{PZH}	3003 (fig. 5)	184	GND	OUT	OUT	OUT						GND							GND	GND	IN	5.0 V	G to A1	5	21	ns
			185	"	"	"	"	OUT					"							"	"	"	"	G to A2	"	"	"
			186	"	"	"	"	"	OUT				"							"	"	"	"	G to A3	"	"	"
			187	"	"	"	"	"	"	OUT			"							"	"	"	"	G to A4	"	"	"
			188	"	"	"	"	"	"	"	OUT		"							"	"	"	"	G to A5	"	"	"
			189	"	"	"	"	"	"	"	"	OUT	"							"	"	"	"	G to A6	"	"	"
			190	"	"	"	"	"	"	"	"	"	"							"	"	"	"	G to A7	"	"	"
			191	"	"	"	"	"	"	"	"	"	"							"	"	"	"	G to A8	"	"	"
	t_{PZL}	3003 (fig. 5)	192	4.5 V	4.5 V	4.5 V	4.5 V						"						OUT	OUT	"	"	G to B1	8	24	"
			193	"	"	"	"	4.5 V					"						"	"	"	"	G to B2	"	"	"
			194	"	"	"	"	"	4.5 V				"						"	"	"	"	G to B3	"	"	"
			195	"	"	"	"	"	"	4.5 V			"					OUT	"	"	"	"	G to B4	"	"	"
			196	"	"	"	"	"	"	"	4.5 V		"					"	"	"	"	"	G to B5	"	"	"
			197	"	"	"	"	"	"	"	"	4.5 V	"					"	"	"	"	"	G to B6	"	"	"
			198	"	"	"	"	"	"	"	"	"	"	OUT				"	"	"	"	"	G to B7	"	"	"
			199	"	"	"	"	"	"	"	"	"	"	"				"	"	"	"	"	G to B8	"	"	"
			200	GND	OUT	OUT	"	"	"	"	"	"	"	"	OUT			"	"	"	4.5 V	"	G to A1	"	"	"
			201	"	"	"	OUT	"	"	"	"	"	"	"	"	OUT		"	"	"	"	"	G to A2	"	"	"
			202	"	"	"	"	OUT	"	"	"	"	"	"	"	"	OUT	"	"	4.5 V	"	"	G to A3	"	"	"
			203	"	"	"	"	"	OUT	"	"	"	"	"	"	"	"	4.5 V	"	"	"	"	G to A4	"	"	"
			204	"	"	"	"	"	"	OUT	"	"	"	"	"	"	"	"	4.5 V	"	"	"	G to A5	"	"	"
			205	"	"	"	"	"	"	"	OUT	"	"	"	"	"	"	"	"	"	"	"	G to A6	"	"	"
			206	"	"	"	"	"	"	"	"	OUT	"	"	"	"	"	"	"	"	"	"	G to A7	"	"	"
			207	"	"	"	"	"	"	"	"	"	"	4.5 V	4.5 V	4.5 V	4.5 V	"	"	"	"	"	G to A8	"	"	"
	t_{PHZ}	3003 (fig. 5)	208	4.5 V	GND	GND	GND						"						OUT	OUT	"	"	G to B1	3	12	"
			209	"	"	"	"	GND					"						"	"	"	"	G to B2	"	"	"
			210	"	"	"	"	"	GND				"						"	"	"	"	G to B3	"	"	"
			211	"	"	"	"	"	"	GND			"					OUT	"	"	"	"	G to B4	"	"	"
			212	"	"	"	"	"	"	"	GND		"					"	"	"	"	"	G to B5	"	"	"
			213	"	"	"	"	"	"	"	"	GND	"					"	"	"	"	"	G to B6	"	"	"
			214	"	"	"	"	"	"	"	"	"	"	OUT				"	"	"	"	"	G to B7	"	"	"
			215	"	"	"	"	"	"	"	"	"	"	"				"	"	"	"	"	G to B8	"	"	"
			216	GND	OUT	OUT	"	"	"	"	"	"	"	"	OUT			"	"	"	GND	"	G to A1	"	"	"
			217	"	"	"	OUT	"	"	"	"	"	"	"	"	OUT		"	"	"	"	"	G to A2	"	"	"
			218	"	"	"	"	OUT	"	"	"	"	"	"	"	"	OUT	"	"	GND	"	"	G to A3	"	"	"
			219	"	"	"	"	"	OUT	"	"	"	"	"	"	"	"	GND	"	"	"	"	G to A4	"	"	"
			220	"	"	"	"	"	"	OUT	"	"	"	"	"	"	"	"	GND	"	"	"	G to A5	"	"	"
			221	"	"	"	"	"	"	"	OUT	"	"	"	"	"	"	"	"	"	"	"	G to A6	"	"	"
			222	"	"	"	"	"	"	"	"	OUT	"	"	"	"	"	"	"	"	"	"	G to A7	"	"	"
			223	"	"	"	"	"	"	"	"	"	"	GND	GND	GND	GND	"	"	"	"	"	G to A8	"	"	"
	t_{PLZ}	3003 (fig. 5)	224	4.5 V	4.5 V	4.5 V	4.5 V						"						OUT	OUT	"	"	G to B1	4	18	"
			225	"	"	"	"	4.5 V					"						"	"	"	"	G to B2	"	"	"
			226	"	"	"	"	"	4.5 V				"						"	"	"	"	G to B3	"	"	"
			227	"	"	"	"	"	"	4.5 V			"					OUT	"	"	"	"	G to B4	"	"	"
			228	"	"	"	"	"	"	"	4.5 V		"					"	"	"	"	"	G to B5	"	"	"
			229	"	"	"	"	"	"	"	"	4.5 V	"					"	"	"	"	"	G to B6	"	"	"
			230	"	"	"	"	"	"	"	"	"	"	OUT				"	"	"	"	"	G to B7	"	"	"
			231	"	"	"	"	"	"	"	"	"	"	"				"	"	"	"	"	G to B8	"	"	"
			232	GND	OUT	OUT	"	"	"	"	"	"	"	"	OUT			"	"	"	4.5 V	"	G to A1	"	"	"
			233	"	"	"	OUT	"	"	"	"	"	"	"	"	OUT		"	"	"	"	"	G to A2	"	"	"
			234	"	"	"	"	OUT	"	"	"	"	"	"	"	"	OUT	"	"	4.5 V	"	"	G to A3	"	"	"
			235	"	"	"	"	"	OUT	"	"	"	"	"	"	"	"	4.5 V	"	"	"	"	G to A4	"	"	"
			236	"	"	"	"	"	"	OUT	"	"	"	"	"	"	"	"	4.5 V	"	"	"	G to A5	"	"	"
			237	"	"	"	"	"	"	"	OUT	"	"	"	"	"	"	"	"	"	"	"	G to A6	"	"	"
			238	"	"	"	"	"	"	"	"	OUT	"	"	"	"	"	"	"	"	"	"	G to A7	"	"	"
			239	"	"	"	"	"	"	"	"	"	"	4.5 V	4.5 V	4.5 V	4.5 V	"	"	"	"	"	G to A8	"	"	"

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,X,2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits		Unit
			Test no.	DIR	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	$\overline{G}$	V_{CC}		Min	Max	
10 T_C = +125°C	t_{PLH1}	3003 (fig. 5)	Same tests and terminal conditions as subgroup 9, except T_C = +125°C.																						3	14	ns
	t_{PHL1}																								2	13	*
	t_{PZH}																								5	25	*
	t_{PZL}																								8	27	*
	t_{PHZ}																								3	15	*
	t_{PLZ}																								4	20	*
11	Same tests, terminal conditions, and limits as subgroup 10 except T_C = -55°C																										

1/ MIL-STD-883, method 3011 shall be used, except the output voltage shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one-half of the true circuit current I_{OS}.

2/ I_O limits for circuit C shall be -30 mA to -110 mA.

3/ A = 3.0 V minimum; B = 0.0 V or GND.

4/ H > 1.5 V; L < 1.5 V.

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TABLE III. Group A inspection for device type 02.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,X,Z	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits		Unit
			Test no.	DIR	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	$\overline{G}$	V_{CC}		Min	Max	
1 T_C = +25°C	I_{OH}		1	2.0 V	2.0 V	2.0 V							GND								5.5 V	0.8 V	4.5 V	B1		100	µA
			2	"																5.5 V				B2			
			3				2.0 V												5.5 V					B3			
			4					2.0 V										5.5 V						B4			
			5						2.0 V								5.5 V							B5			
			6							2.0 V						5.5 V								B6			
			7								2.0 V				5.5 V									B7			
			8									2.0 V		5.5 V										B8			
			9	0.8 V	5.5 V																2.0 V			A1			
			10	"		5.5 V														2.0 V				A2			
			11				5.5 V												2.0 V					A3			
			12					5.5 V										2.8 V						A4			
			13						5.5 V								2.0 V							A5			
			14							5.5 V						2.0 V								A6			
			15								5.5 V				2.0 V									A7			
			16									5.5 V		2.0 V										A8			
	V_{OL}	3006	17	2.0 V	0.8 V															12 mA	12 mA			B1		0.4	V
			18	"		0.8 V													12 mA					B2			
			19				0.8 V											12 mA						B3			
			20					0.8 V									12 mA							B4			
			21						0.8 V							12 mA								B5			
			22							0.8 V					12 mA									B6			
			23								0.8 V			12 mA										B7			
			24		-							0.8 V												B8			
			25	0.8 V	12 mA																0.8 V			A1			
			26	"		12 mA														0.8 V				A2			
			27				12 mA												0.8 V					A3			
			28					12 mA										0.8 V						A4			
			29						12 mA								0.8 V							A5			
			30							12 mA						0.8 V								A6			
			31								12 mA				0.8 V									A7			
			32									12 mA		0.8 V										A8			
	V_{IC}		33	-18 mA																				DIR		1.5	
			34		-18 mA																			A1			
			35			-18 mA																		A2			
			36				-18 mA																	A3			
			37					-18 mA																A4			
			38						-18 mA															A5			
			39			0				-18 mA														A6			
			40								-18 mA													A7			
			41									-18 mA												A8			
			42											-18 mA										B8			
			43												-18 mA									B7			
			44													-18 mA								B6			
			45														-18 mA							B5			
			46															-18 mA						B4			
			47																-18 mA					B3			
			48																	-18 mA				B2			
			49																		-18 mA			B1			
			50																			-18 mA		$\overline{G}$			
	I_{IL}	3009	51	0.4 V																			5.5 V	DIR	0	-100	µA
			52		0.4 V																			A1			
			53			0.4 V																		A2			
			54				0.4 V																	A3			
			55					0.4 V																A4			
			56						0.4 V															A5			
			57							0.4 V														A6			
			58								0.4 V													A7			
			59									0.4 V												A8			
			60											0.4 V										B8			
			61												0.4 V									B7			
			62													0.4 V								B6			
			63														0.4 V							B5			
			64															0.4 V						B4			
			65																0.4 V					B3			
			66																	0.4 V				B2			

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,X,Z: Test no.	1: DIR	2: A1	3: A2	4: A3	5: A4	6: A5	7: A6	8: A7	9: A8	10: GND	11: B8	12: B7	13: B6	14: B5	15: B4	16: B3	17: B2	18: B1	19: $\overline{G}$	20: V_{CC}	Measured terminal	Test limits: Min	Test limits: Max	Unit
1 T_C = +25°C	I_{IL}	3009	67										GND								0.4 V	5.5 V	B1	0	-100	µA
	I_{IL}	3009	68										"								0.4 V	"	$\overline{G}$	0	-100	"
	I_{IH1}	3010	69	2.7 V									"									"	DIR		20	"
			70		2.7 V								"									"	A1		"	"
			71			2.7 V							"									"	A2		"	"
			72				2.7 V						"									"	A3		"	"
			73					2.7 V					"									"	A4		"	"
			74						2.7 V				"									"	A5		"	"
			75							2.7 V			"									"	A6		"	"
			76								2.7 V		"									"	A7		"	"
			77									2.7 V	"									"	A8		"	"
			78										"	2.7 V								"	B8		"	"
			79										"		2.7 V							"	B7		"	"
			80										"			2.7 V						"	B6		"	"
			81										"				2.7 V					"	B5		"	"
			82										"					2.7 V				"	B4		"	"
			83										"						2.7 V			"	B3		"	"
			84										"							2.7 V		"	B2		"	"
			85										"								2.7 V	"	B1		"	"
			86										"									"	$\overline{G}$		"	"
	I_{IH2}	3010	87	7.0 V									"									"	DIR		100	"
			88		7.0 V								"									"	A1		"	"
			89			7.0 V							"									"	A2		"	"
			90				7.0 V						"									"	A3		"	"
			91					7.0 V					"									"	A4		"	"
			92						7.0 V				"									"	A5		"	"
			93							7.0 V			"									"	A6		"	"
			94								7.0 V		"									"	A7		"	"
			95									7.0 V	"									"	A8		"	"
			96										"	7.0 V								"	B8		"	"
			97										"		7.0 V							"	B7		"	"
			98										"			7.0 V						"	B6		"	"
			99										"				7.0 V					"	B5		"	"
			100										"					7.0 V				"	B4		"	"
			101										"						7.0 V			"	B3		"	"
			102										"							7.0 V		"	B2		"	"
			103										"								7.0 V	"	B1		"	"
			104										"									"	$\overline{G}$		"	"
	I_{CCH}	3005	105	GND									"	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	GND	"	V_{CC}		37	mA
	I_{CCL}	3005	106	GND									"	GND	GND	GND	GND	GND	GND	GND	GND	GND	"	V_{CC}		45	mA
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C and omit V_{IC} tests.																									
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and omit V_{IC} tests.																									
7 T_C = +25°C	Truth table tests	3014	107	B 1/	B	B	B	B	B	B	R	R	R	GND	L	L	L	L	L	L	L	L	B	4.5 V	See 2/			
			108	B	A	A	A	A	A	A	A	A	A	"	H	H	H	H	H	H	H	H	B	"				
			109	A 1/	L	L	L	L	L	L	L	L	L	"	B	B	B	B	B	B	B	B	B	"				
			110	A	H	H	H	H	H	H	H	H	H	"	A	A	A	A	A	A	A	A	B	"				
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																									

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,X,2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits		Unit
			Test no.	DIR	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	$\overline{G}$	V_{CC}		Min	Max	
9 T_C = +25°C	t_{PLH1}	3003 (fig. 5)	111	GND	IN	IN	IN	IN	IN	IN	IN	IN	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	5.0 V	A1 to B1	10	30	ns
			112	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A2 to B2	"	"	"
			113	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A3 to B3	"	"	"
			114	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A4 to B4	"	"	"
			115	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A5 to B5	"	"	"
			116	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A6 to B6	"	"	"
			117	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A7 to B7	"	"	"
			118	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A8 to B8	"	"	"
			119	4.5 V	OUT	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	IN	"	"	B1 to A1	"	"	"
			120	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B2 to A2	"	"	"
			121	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B3 to A3	"	"	"
			122	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B4 to A4	"	"	"
			123	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B5 to A5	"	"	"
			124	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B6 to A6	"	"	"
			125	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B7 to A7	"	"	"
			126	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B8 to A8	"	"	"
	t_{PHL1}	3003 (fig. 5)	127	GND	IN	IN	IN	IN	IN	IN	IN	IN	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	"	A1 to B1	5	19	"
			128	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A2 to B2	"	"	"
			129	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A3 to B3	"	"	"
			130	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A4 to B4	"	"	"
			131	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A5 to B5	"	"	"
			132	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A6 to B6	"	"	"
			133	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A7 to B7	"	"	"
			134	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A8 to B8	"	"	"
			135	4.5 V	OUT	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	IN	"	"	B1 to A1	"	"	"
			136	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B2 to A2	"	"	"
			137	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B3 to A3	"	"	"
			138	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B4 to A4	"	"	"
			139	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B5 to A5	"	"	"
			140	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B6 to A6	"	"	"
			141	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B7 to A7	"	"	"
			142	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B8 to A8	"	"	"
	t_{PLH2}	3003 (fig. 5)	143	IN	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	"	DIR to A1	13	35	"
			144	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DIR to A2	"	"	"
			145	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DIR to A3	"	"	"
			146	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DIR to A4	"	"	"
			147	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DIR to A5	"	"	"
			148	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DIR to A6	"	"	"
			149	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DIR to A7	"	"	"
			150	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DIR to A8	"	"	"
			151	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DIR to B8	"	"	"
			152	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DIR to B7	"	"	"
			153	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DIR to B6	"	"	"
			154	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DIR to B5	"	"	"
			155	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DIR to B4	"	"	"
			156	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DIR to B3	"	"	"
			157	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DIR to B2	"	"	"
			158	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DIR to B1	"	"	"
	t_{PHL2}	3003 (fig. 5)	159	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	"	DIR to A1	"	"	"
			160	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DIR to A2	"	"	"
			161	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DIR to A3	"	"	"
			162	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DIR to A4	"	"	"
			163	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DIR to A5	"	"	"
			164	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DIR to A6	"	"	"
			165	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DIR to A7	"	"	"
			166	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DIR to A8	"	"	"
			167	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DIR to B8	"	"	"
			168	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DIR to B7	"	"	"
			169	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DIR to B6	"	"	"
			170	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DIR to B5	"	"	"
			171	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DIR to B4	"	"	"
			172	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DIR to B3	"	"	"
			173	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DIR to B2	"	"	"
			174	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	DIR to B1	"	"	"

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,X,2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits		Unit
			Test no.	DIR	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	$\overline{G}$	V_{CC}		Min	Max	
9 T_C = +25°C	t_{PLH2}	3003 (fig. 5)	175		OUT								GND									IN	5.0 V	$\overline{G}$ to A1	13	35	ns
			176			OUT							"									"	"	$\overline{G}$ to A2	"	"	"
			177				OUT						"									"	"	$\overline{G}$ to A3	"	"	"
			178					OUT					"									"	"	$\overline{G}$ to A4	"	"	"
			179						OUT				"									"	"	$\overline{G}$ to A5	"	"	"
			180							OUT			"									"	"	$\overline{G}$ to A6	"	"	"
			181								OUT		"									"	"	$\overline{G}$ to A7	"	"	"
			182									OUT	"									"	"	$\overline{G}$ to A8	"	"	"
			183										"	OUT								"	"	$\overline{G}$ to B8	"	"	"
			184										"		OUT							"	"	$\overline{G}$ to B7	"	"	"
			185										"			OUT						"	"	$\overline{G}$ to B6	"	"	"
			186										"				OUT					"	"	$\overline{G}$ to B5	"	"	"
			187										"					OUT				"	"	$\overline{G}$ to B4	"	"	"
			188										"						OUT			"	"	$\overline{G}$ to B3	"	"	"
			189										"							OUT		"	"	$\overline{G}$ to B2	"	"	"
			190										"								OUT	"	"	$\overline{G}$ to B1	"	"	"
	t_{PHL3}	3003 (fig. 5)	191		OUT								"										"	$\overline{G}$ to A1	"	"	"
			192			OUT							"										"	$\overline{G}$ to A2	"	"	"
			193				OUT						"										"	$\overline{G}$ to A3	"	"	"
			194					OUT					"										"	$\overline{G}$ to A4	"	"	"
			195						OUT				"										"	$\overline{G}$ to A5	"	"	"
			196							OUT			"										"	$\overline{G}$ to A6	"	"	"
			197								OUT		"										"	$\overline{G}$ to A7	"	"	"
			198									OUT	"										"	$\overline{G}$ to A8	"	"	"
			199										"	OUT									"	$\overline{G}$ to B8	"	"	"
			200										"		OUT								"	$\overline{G}$ to B7	"	"	"
			201										"			OUT							"	$\overline{G}$ to B6	"	"	"
			202										"				OUT						"	$\overline{G}$ to B5	"	"	"
			203										"					OUT					"	$\overline{G}$ to B4	"	"	"
			204										"						OUT				"	$\overline{G}$ to B3	"	"	"
			205										"							OUT			"	$\overline{G}$ to B2	"	"	"
			206										"								OUT		"	$\overline{G}$ to B1	"	"	"
10 T_C = +125°C	t_{PLH1} t_{PHL1} t_{PLH2} t_{PHL2}	3003 (fig. 5)	Same tests and terminal conditions as subgroup 9, except T_C = +125°C.																						10 5 13 13	37 23 40 40	" " " "
11	Same tests, terminal conditions, and limits, as subgroup 10, except T_C = -55°C																										

1/ A = 3.0 V minimum, B = 0.0 V or GND.

2/ H > 1.5 V; L < 1.5 V.

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TABLE III. Group A inspection for device type 03.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,X,2 Test no.	1 DIR	2 A1	3 A2	4 A3	5 A4	6 A5	7 A6	8 A7	9 A8	10 GND	11 B8	12 B7	13 B6	14 B5	15 B4	16 B3	17 B2	18 B1	19 $\overline{G}$	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	I_{OH}		1	2.0 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	GND	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	0.8 V	4.5 V	B1		100	µA
			2	"									"											B2		"	"
			3	"									"											B3		"	"
			4	"									"											B4		"	"
			5	"									"											B5		"	"
			6	"									"											B6		"	"
			7	"									"											B7		"	"
			8	"									"											B8		"	"
			9	0.8 V	5.5 V								"											A1		"	"
			10	"		5.5 V							"											A2		"	"
			11	"			5.5 V						"											A3		"	"
			12	"				5.5 V					"											A4		"	"
			13	"					5.5 V				"											A5		"	"
			14	"						5.5 V			"											A6		"	"
			15	"							5.5 V		"											A7		"	"
			16	"								5.5 V	"											A8		"	"
	V_{OL}	3006	17	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	"	12 mA	12 mA	12 mA	12 mA	12 mA	12 mA	12 mA	12 mA	"	"	B1		0.4	V
			18	"									"											B2		"	"
			19	"									"											B3		"	"
			20	"									"											B4		"	"
			21	"									"											B5		"	"
			22	"									"											B6		"	"
			23	"									"											B7		"	"
			24	"									"											B8		"	"
			25	0.8 V	12 mA								"											A1		"	"
			26	"		12 mA							"											A2		"	"
			27	"			12 mA						"											A3		"	"
			28	"				12 mA					"											A4		"	"
			29	"					12 mA				"											A5		"	"
			30	"						12 mA			"											A6		"	"
			31	"							12 mA		"											A7		"	"
			32	"								12 mA	"	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	"	"	A8		"	"
	V_{IC}		33	-18 mA									"											DIR		1.5	"
			34	"	-18 mA								"											A1		"	"
			35	"		-18 mA							"											A2		"	"
			36	"			-18 mA						"											A3		"	"
			37	"				-18 mA					"											A4		"	"
			38	"					-18 mA				"											A5		"	"
			39	"						-18 mA			"											A6		"	"
			40	"							-18 mA		"											A7		"	"
			41	"								-18 mA	"	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	"	"	A8		"	"
			42	"									"											B8		"	"
			43	"									"											B7		"	"
			44	"									"											B6		"	"
			45	"									"											B5		"	"
			46	"									"											B4		"	"
			47	"									"											B3		"	"
			48	"									"											B2		"	"
			49	"									"											B1		"	"
			50	"									"									-18 mA		$\overline{G}$		"	"
	I_{IL}	3009	51	0.4 V									"										5.5 V	DIR	0	-100	µA
			52	"	0.4 V								"										"	A1	"	"	"
			53	"		0.4 V							"										"	A2	"	"	"
			54	"			0.4 V						"										"	A3	"	"	"
			55	"				0.4 V					"										"	A4	"	"	"
			56	"					0.4 V				"										"	A5	"	"	"
			57	"						0.4 V			"										"	A6	"	"	"
			58	"							0.4 V		"										"	A7	"	"	"
			59	"								0.4 V	"	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	"	"	A8	"	"	"
			60	"									"										"	B8	"	"	"
			61	"									"										"	B7	"	"	"
			62	"									"										"	B6	"	"	"
			63	"									"										"	B5	"	"	"
			64	"									"										"	B4	"	"	"
			65	"									"										"	B3	"	"	"
			66	"									"										"	B2	"	"	"

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,X,2 Test no.	1 DIR	2 A1	3 A2	4 A3	5 A4	6 A5	7 A6	8 A7	9 A8	10 GND	11 B8	12 B7	13 B6	14 B5	15 B4	16 B3	17 B2	18 B1	19 $\overline{G}$	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	I_{IL}	3009	67										GND								0.4 V	5.5 V	B1	0	-100	µA
	I_{IL}	3009	68										"								0.4 V	"	$\overline{G}$	0	-100	"
	I_{IH1}	3010	69	2.7 V									"									"	DIR		20	"
			70		2.7 V								"									"	A1		"	"
			71			2.7 V							"									"	A2		"	"
			72				2.7 V						"									"	A3		"	"
			73					2.7 V					"									"	A4		"	"
			74						2.7 V				"									"	A5		"	"
			75							2.7 V			"									"	A6		"	"
			76								2.7 V		"									"	A7		"	"
			77									2.7 V	"									"	A8		"	"
			78										"	2.7 V								"	B8		"	"
			79										"		2.7 V							"	B7		"	"
			80										"			2.7 V						"	B6		"	"
			81										"				2.7 V					"	B5		"	"
			82										"					2.7 V				"	B4		"	"
			83										"						2.7 V			"	B3		"	"
			84										"							2.7 V		"	B2		"	"
			85										"								2.7 V	"	B1		"	"
			86										"									"	$\overline{G}$		"	"
	I_{IH2}	3010	87	7.0 V									"									"	DIR		100	"
			88		7.0 V								"									"	A1		"	"
			89			7.0 V							"									"	A2		"	"
			90				7.0 V						"									"	A3		"	"
			91					7.0 V					"									"	A4		"	"
			92						7.0 V				"									"	A5		"	"
			93							7.0 V			"									"	A6		"	"
			94								7.0 V		"									"	A7		"	"
			95									7.0 V	"									"	A8		"	"
			96										"	7.0 V								"	B8		"	"
			97										"		7.0 V							"	B7		"	"
			98										"			7.0 V						"	B6		"	"
			99										"				7.0 V					"	B5		"	"
			100										"					7.0 V				"	B4		"	"
			101										"						7.0 V			"	B3		"	"
			102										"							7.0 V		"	B2		"	"
			103										"								7.0 V	"	B1		"	"
			104										"									"	$\overline{G}$		"	"
	I_{CCH}	3005	105	GND									"	GND	GND	GND	GND	GND	GND	GND	GND	5.5 V	V_{CC}		15	mA
	I_{CCL}	3005	106	GND									"	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	V_{CC}		20	mA
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C and omit V_{IC} tests.																									
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and omit V_{IC} tests.																									
7 T_C = +25°C	Truth table tests	3014	107	B 1/ B A 1/ A	A B L H	A B L H	A B L H	A B L H	A B L H	A B L H	A B H H	A B H H	GND " "	L H ^ B	L H ^ B	L H ^ B	L H ^ B	L H ^ B	L H ^ B	L H ^ B	L H ^ B	B B " B	4.5 V " "	See 2/			
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																									

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,X,Z	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits		Unit
			Test no.	DIR	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	$\overline{G}$	V_{CC}		Min	Max	
T_C = +25°C [9]	t_{PLH1}	3003 (fig. 5)	111	GND	IN	IN	IN	IN	IN	IN	IN	IN	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	5.0 V	A1 to B1	15	41	ns
			112	"									"									"	"	A2 to B2	"	"	
			113	"									"									"	"	A3 to B3	"	"	
			114	"									"									"	"	A4 to B4	"	"	
			115	"									"									"	"	A5 to B5	"	"	
			116	"									"									"	"	A6 to B6	"	"	
			117	"									"									"	"	A7 to B7	"	"	
			118	"									"									"	"	A8 to B8	"	"	
			119	4.5 V	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	IN	IN	IN	IN	IN	IN	IN	IN	"	"	B1 to A1	"	"	
			120	"									"									"	"	B2 to A2	"	"	
			121	"									"									"	"	B3 to A3	"	"	
			122	"									"									"	"	B4 to A4	"	"	
			123	"									"									"	"	B5 to A5	"	"	
			124	"									"									"	"	B6 to A6	"	"	
			125	"									"									"	"	B7 to A7	"	"	
			126	"									"									"	"	B8 to A8	"	"	
	t_{PHL1}	3003 (fig. 5)	127	GND	IN	IN	IN	IN	IN	IN	IN	IN	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	"	A1 to B1	5	22	"
			128	"									"									"	"	A2 to B2	"	"	
			129	"									"									"	"	A3 to B3	"	"	
			130	"									"									"	"	A4 to B4	"	"	
			131	"									"									"	"	A5 to B5	"	"	
			132	"									"									"	"	A6 to B6	"	"	
			133	"									"									"	"	A7 to B7	"	"	
			134	"									"									"	"	A8 to B8	"	"	
			135	4.5 V	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	IN	IN	IN	IN	IN	IN	IN	IN	"	"	B1 to A1	"	"	
			136	"									"									"	"	B2 to A2	"	"	
			137	"									"									"	"	B3 to A3	"	"	
			138	"									"									"	"	B4 to A4	"	"	
			139	"									"									"	"	B5 to A5	"	"	
			140	"									"									"	"	B6 to A6	"	"	
			141	"									"									"	"	B7 to A7	"	"	
			142	"									"									"	"	B8 to A8	"	"	
	t_{PLH2}	3003 (fig. 5)	143	IN	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT		"	DIR to A1	15	40	"
			144	"									"										"	DIR to A2	"	"	
			145	"									"										"	DIR to A3	"	"	
			146	"									"										"	DIR to A4	"	"	
			147	"									"										"	DIR to A5	"	"	
			148	"									"										"	DIR to A6	"	"	
			149	"									"										"	DIR to A7	"	"	
			150	"									"										"	DIR to A8	"	"	
			151	"									"										"	DIR to B8	"	"	
			152	"									"										"	DIR to B7	"	"	
			153	"									"										"	DIR to B6	"	"	
			154	"									"										"	DIR to B5	"	"	
			155	"									"										"	DIR to B4	"	"	
			156	"									"										"	DIR to B3	"	"	
			157	"									"										"	DIR to B2	"	"	
			158	"									"										"	DIR to B1	"	"	
	t_{PHL2}	3003 (fig. 5)	159	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT		"	DIR to A1	15	45	"
			160	"									"										"	DIR to A2	"	"	
			161	"									"										"	DIR to A3	"	"	
			162	"									"										"	DIR to A4	"	"	
			163	"									"										"	DIR to A5	"	"	
			164	"									"										"	DIR to A6	"	"	
			165	"									"										"	DIR to A7	"	"	
			166	"									"										"	DIR to A8	"	"	
			167	"									"										"	DIR to B8	"	"	
			168	"									"										"	DIR to B7	"	"	
			169	"									"										"	DIR to B6	"	"	
			170	"									"										"	DIR to B5	"	"	
			171	"									"										"	DIR to B4	"	"	
			172	"									"										"	DIR to B3	"	"	
			173	"									"										"	DIR to B2	"	"	
			174	"									"										"	DIR to B1	"	"	

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,X,2 Test no.	1 DIR	2 A1	3 A2	4 A3	5 A4	6 A5	7 A6	8 A7	9 A8	10 GND	11 B8	12 B7	13 B6	14 B5	15 B4	16 B3	17 B2	18 B1	19 $\overline{G}$	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
9 T_C = +25°C	t_{PLH2}	3003 (fig. 5)	175		OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND									IN	5.0 V	$\overline{G}$ to A1	15	40	ns
			176										"									"	"	$\overline{G}$ to A2	"	"	"
			177										"									"	"	$\overline{G}$ to A3	"	"	"
			178										"									"	"	$\overline{G}$ to A4	"	"	"
			179										"									"	"	$\overline{G}$ to A5	"	"	"
			180										"									"	"	$\overline{G}$ to A6	"	"	"
			181										"									"	"	$\overline{G}$ to A7	"	"	"
			182										"	OUT								"	"	$\overline{G}$ to A8	"	"	"
			183										"		OUT							"	"	$\overline{G}$ to B8	"	"	"
			184										"			OUT						"	"	$\overline{G}$ to B7	"	"	"
			185										"				OUT					"	"	$\overline{G}$ to B6	"	"	"
			186										"					OUT				"	"	$\overline{G}$ to B5	"	"	"
			187										"						OUT			"	"	$\overline{G}$ to B4	"	"	"
			188										"							OUT		"	"	$\overline{G}$ to B3	"	"	"
			189										"								OUT	"	"	$\overline{G}$ to B2	"	"	"
			190										"									"	"	$\overline{G}$ to B1	"	"	"
	t_{PHL3}	3003 (fig. 5)	191		OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"										"	$\overline{G}$ to A1	"	45	"
			192										"										"	$\overline{G}$ to A2	"	"	"
			193										"										"	$\overline{G}$ to A3	"	"	"
			194										"										"	$\overline{G}$ to A4	"	"	"
			195										"										"	$\overline{G}$ to A5	"	"	"
			196										"										"	$\overline{G}$ to A6	"	"	"
			197										"										"	$\overline{G}$ to A7	"	"	"
			198										"										"	$\overline{G}$ to A8	"	"	"
			199										"	OUT									"	$\overline{G}$ to B8	"	"	"
			200										"		OUT								"	$\overline{G}$ to B7	"	"	"
			201										"			OUT							"	$\overline{G}$ to B6	"	"	"
			202										"				OUT						"	$\overline{G}$ to B5	"	"	"
			203										"					OUT					"	$\overline{G}$ to B4	"	"	"
			204										"						OUT				"	$\overline{G}$ to B3	"	"	"
			205										"							OUT			"	$\overline{G}$ to B2	"	"	"
			206										"								OUT		"	$\overline{G}$ to B1	"	"	"
10 T_C = +125°C	t_{PLH1} t_{PHL1} t_{PLH2} t_{PHL2}	3003 (fig. 5)	Same tests and terminal conditions as subgroup 9, except T_C = +125°C.																						15 5 15 15	50 25 45 50	" " " "
11	Same tests, terminal conditions, and limits, as subgroup 10, except T_C = -55°C																										

1/ A = 3.0 V minimum, B = 0.0 V or GND.

2/ H > 1.5 V; L < 1.5 V.

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TABLE III. Group A inspection for device type 04.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,X,Z	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
			Test no.	DIR	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	$\overline{G}$	V_{CC}				
1 T_C = +25°C	V_{OH1}	3006	1	2.0 V	0.8 V	0.8 V							GND								-3 mA	0.8 V	4.5 V	B1	2.4		V
			2	"	"		0.8 V						"						-3 mA	-3 mA	"	"	"	B2	"		"
			3	"	"			0.8 V					"					-3 mA			"	"	"	"	B3	"		"
			4	"	"				0.8 V				"				-3 mA				"	"	"	"	B4	"		"
			5	"	"					0.8 V			"			-3 mA					"	"	"	"	B5	"		"
			6	"	"						0.8 V		"		-3 mA						"	"	"	"	B6	"		"
			7	"	"							0.8 V	"	-3 mA							"	"	"	"	B7	"		"
			8	"	"								"								"	"	"	"	B8	"		"
			9	0.8 V	-3 mA								"								2.0 V	"	"	"	A1	"		"
			10	"		-3 mA							"							2.0 V	"	"	"	"	A2	"		"
			11	"			-3 mA						"						2.0 V		"	"	"	"	A3	"		"
			12	"				-3 mA					"					2.0 V			"	"	"	"	A4	"		"
			13	"					-3 mA				"				2.0 V				"	"	"	"	A5	"		"
			14	"						-3 mA			"			2.0 V					"	"	"	"	A6	"		"
			15	"							-3 mA		"		2.0 V						"	"	"	"	A7	"		"
			16	"								-3 mA	"	2.0 V							"	"	"	"	A8	"		"
	V_{OH2}	3006	17	2.0 V	0.8 V	0.8 V							"								-12 mA	-12 mA	"	"	B1	2.0		"
			18	"	"		0.8 V						"						-12 mA	-12 mA	"	"	"	"	B2	"		"
			19	"	"			0.8 V					"					-12 mA			"	"	"	"	B3	"		"
			20	"	"				0.8 V				"				-12 mA				"	"	"	"	B4	"		"
			21	"	"					0.8 V			"			-12 mA					"	"	"	"	B5	"		"
			22	"	"						0.8 V		"		-12 mA						"	"	"	"	B6	"		"
			23	"	"							0.8 V	"	-12 mA							"	"	"	"	B7	"		"
			24	"	"								"								"	"	"	"	B8	"		"
			25	0.8 V	-12 mA								"								2.0 V	"	"	"	A1	"		"
			26	"		-12 mA							"							2.0 V	"	"	"	"	A2	"		"
			27	"			-12 mA						"						2.0 V		"	"	"	"	A3	"		"
			28	"				-12 mA					"					2.0 V			"	"	"	"	A4	"		"
			29	"					-12 mA				"				2.0 V				"	"	"	"	A5	"		"
			30	"						-12 mA			"			2.0 V					"	"	"	"	A6	"		"
			31	"							-12 mA		"		2.0 V						"	"	"	"	A7	"		"
			32	"								-12 mA	"	2.0 V							"	"	"	"	A8	"		"
	V_{OL}	3007	33	2.0 V	2.0 V	2.0 V							"								12 mA	12 mA	"	"	B1		0.4	"
			34	"	"		2.0 V						"						12 mA	12 mA	"	"	"	"	B2		"	"
			35	"	"			2.0 V					"					12 mA			"	"	"	"	B3		"	"
			36	"	"				2.0 V				"				12 mA				"	"	"	"	B4		"	"
			37	"	"					2.0 V			"			12 mA					"	"	"	"	B5		"	"
			38	"	"						2.0 V		"		12 mA						"	"	"	"	B6		"	"
			39	"	"							2.0 V	"	12 mA							"	"	"	"	B7		"	"
			40	"	"								"								"	"	"	"	B8		"	"
			41	0.8 V	12 mA								"								0.8 V	"	"	"	A1		"	"
			42	"		12 mA							"							0.8 V	"	"	"	"	A2		"	"
			43	"			12 mA						"						0.8 V		"	"	"	"	A3		"	"
			44	"				12 mA					"					0.8 V			"	"	"	"	A4		"	"
			45	"					12 mA				"				0.8 V				"	"	"	"	A5		"	"
			46	"						12 mA			"			0.8 V					"	"	"	"	A6		"	"
			47	"							12 mA		"		0.8 V						"	"	"	"	A7		"	"
			48	"								12 mA	"	0.8 V							"	"	"	"	A8		"	"
	V_{IC}		49	-18 mA									"										4.5 V	DIR		1.5	"
			50	"	-18 mA								"										"	A1		"	"
			51	"	"	-18 mA							"										"	A2		"	"
			52	"	"		-18 mA						"										"	A3		"	"
			53	"	"			-18 mA					"										"	A4		"	"
			54	"	"				-18 mA				"										"	A5		"	"
			55	"	"					-18 mA			"										"	A6		"	"
			56	"	"						-18 mA		"										"	A7		"	"
			57	"	"							-18 mA	"										"	A8		"	"
			58	"	"								"										"	B8		"	"
			59	"	"								"										"	B7		"	"
			60	"	"								"			-18 mA							"	B6		"	"
			61	"	"								"				-18 mA						"	B5		"	"
			62	"	"								"					-18 mA					"	B4		"	"
			63	"	"								"						-18 mA				"	B3		"	"
			64	"	"								"							-18 mA			"	B2		"	"
			65	"	"								"								-18 mA		"	B1		"	"
			66	"	"								"									-18 mA	"	$\overline{G}$		"	"

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,X,2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				DIR	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	$\overline{G}$	V_{CC}				
1 T_C = +25°C	I_{IL}	3009	67	0.4 V									GND									0.4 V	5.5 V	DIR	0	-100	µA
	I_{IL}	3009	68	"									"									"	"	$\overline{G}$	"	"	"
	I_{IL}/ I_{OZL}		69		0.4 V								"									4.5 V	"	A1	"	"	"
			70			0.4 V							"									"	"	A2	"	"	"
			71				0.4 V						"									"	"	A3	"	"	"
			72					0.4 V					"									"	"	A4	"	"	"
			73						0.4 V				"									"	"	A5	"	"	"
			74							0.4 V			"									"	"	A6	"	"	"
			75								0.4 V		"									"	"	A7	"	"	"
			76									0.4 V	"									"	"	A8	"	"	"
			77										"	0.4 V								"	"	B8	"	"	"
			78										"		0.4 V							"	"	B7	"	"	"
			79										"			0.4 V						"	"	B6	"	"	"
			80										"				0.4 V					"	"	B5	"	"	"
			81										"					0.4 V				"	"	B4	"	"	"
			82										"						0.4 V			"	"	B3	"	"	"
			83										"							0.4 V		"	"	B2	"	"	"
			84										"								0.4 V	"	"	B1	"	"	"
	I_{IH1}	3010	85	2.7 V									"									2.7 V	"	DIR		20	"
	I_{IH1}	3010	86	"									"									"	"	$\overline{G}$		"	"
	I_{IH1}/ I_{OZH}		87		2.7 V								"									4.5 V	"	A1		"	"
			88			2.7 V							"									"	"	A2		"	"
			89				2.7 V						"									"	"	A3		"	"
			90					2.7 V					"									"	"	A4		"	"
			91						2.7 V				"									"	"	A5		"	"
			92							2.7 V			"									"	"	A6		"	"
			93								2.7 V		"									"	"	A7		"	"
			94									2.7 V	"									"	"	A8		"	"
			95										"	2.7 V								"	"	B8		"	"
			96										"		2.7 V							"	"	B7		"	"
			97										"			2.7 V						"	"	B6		"	"
			98										"				2.7 V					"	"	B5		"	"
			99										"					2.7 V				"	"	B4		"	"
			100										"						2.7 V			"	"	B3		"	"
			101										"							2.7 V		"	"	B2		"	"
			102										"								2.7 V	"	"	B1		"	"
	I_{IH2}	3010	103		5.5 V								"									"	"	A1		100	"
			104			5.5 V							"									"	"	A2		"	"
			105				5.5 V						"									"	"	A3		"	"
			106					5.5 V					"									"	"	A4		"	"
			107						5.5 V				"									"	"	A5		"	"
			108							5.5 V			"									"	"	A6		"	"
			109								5.5 V		"									"	"	A7		"	"
			110									5.5 V	"									"	"	A8		"	"
			111										"	5.5 V								"	"	B8		"	"
			112										"		5.5 V							"	"	B7		"	"
			113										"			5.5 V						"	"	B6		"	"
			114										"				5.5 V					"	"	B5		"	"
			115										"					5.5 V				"	"	B4		"	"
			116										"						5.5 V			"	"	B3		"	"
			117										"							5.5 V		"	"	B2		"	"
			118										"								5.5 V	"	"	B1		"	"
	I_{IH3}	3010	119	7.0 V									"									7.0 V	"	DIR		"	"
		3010	120	"									"									"	"	$\overline{G}$		"	"
	I_O 1/	3011	121	4.5 V	GND								"								2.25 V	GND	"	B1	27 -30	27 -112	mA
			122	"		GND							"							2.25 V		"	"	B2	"	"	"
			123	"			GND						"						2.25 V			"	"	B3	"	"	"
			124	"				GND					"					2.25 V				"	"	B4	"	"	"
			125	"					GND				"				2.25 V					"	"	B5	"	"	"
			126	"						GND			"			2.25 V						"	"	B6	"	"	"
			127	"							GND		"		2.25 V							"	"	B7	"	"	"
			128	"								GND	"	2.25 V								"	"	B8	"	"	"

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,X,Z Test no.	1 DIR	2 A1	3 A2	4 A3	5 A4	6 A5	7 A6	8 A7	9 A8	10 GND	11 B8	12 B7	13 B6	14 B5	15 B4	16 B3	17 B2	18 B1	19 $\overline{G}$	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	I_O 1/	3011	129	GND	2.25 V	2.25 V							GND									GND	5.5 V	A1	2/ -30	2/ -112	mA
			130	"			2.25 V						"							4.5 V		"	"	A2	"	"	"
			131	"				2.25 V					"					4.5 V	4.5 V				"	"	A3	"	"	"
			132	"					2.25 V				"				4.5 V					"	"	A4	"	"	"
			133	"						2.25 V			"			4.5 V						"	"	A5	"	"	"
			134	"							2.25 V		"	4.5 V								"	"	A6	"	"	"
			135	"								2.25 V	"		4.5 V							"	"	A7	"	"	"
			136	"									"									"	"	A8	"	"	"
	I_{CCH}	3005	137	5.5 V	GND	GND	GND	GND	GND	GND	GND	GND	"									"	"	V_{CC}		41	"
	I_{CCL}	3005	138	"	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"									"	"	"		51	"
	I_{CCZ}	3005	139	"									"									5.5 V	"	"		53	"
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C and omit V_{IC} tests.																										
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and omit V_{IC} tests.																										
7 T_C = +25°C	Truth table tests	3014	140	A 3/	A 3/	A	A	A	A	A	A	A	GND	L	L	L	L	L	L	L	L	B	4.5 V	See 4/			
			141	A	B 3/	B	B	B	B	B	B	B	"	H	H	H	H	H	H	H	H	"	"				
			142	B	L	L	L	L	L	L	L	L	"	B	B	B	B	B	B	B	B	"	"				
			143	B	H	H	H	H	H	H	H	H	"	A	A	A	A	A	A	A	A	"	"				
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																										
9 T_C = +25°C	t_{PLH1}	3003 (fig. 5)	144	4.5 V	IN	IN	IN	IN	IN	IN	IN	IN	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	5.0 V	A1 to B1	3	13	ns
			145	"									"									"	"	A2 to B2	"	"	"
			146	"									"									"	"	A3 to B3	"	"	"
			147	"									"									"	"	A4 to B4	"	"	"
			148	"									"									"	"	A5 to B5	"	"	"
			149	"									"									"	"	A6 to B6	"	"	"
			150	"									"									"	"	A7 to B7	"	"	"
			151	"									"									"	"	A8 to B8	"	"	"
			152	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	IN	"	"	B1 to A1	"	"	"
			153	"									"									"	"	B2 to A2	"	"	"
			154	"									"									"	"	B3 to A3	"	"	"
			155	"									"									"	"	B4 to A4	"	"	"
			156	"									"									"	"	B5 to A5	"	"	"
			157	"									"									"	"	B6 to A6	"	"	"
			158	"									"									"	"	B7 to A7	"	"	"
			159	"									"									"	"	B8 to A8	"	"	"
	t_{PHL1}	3003 (fig. 5)	160	4.5 V	IN	IN	IN	IN	IN	IN	IN	IN	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	"	A1 to B1	2	11	"
			161	"									"									"	"	A2 to B2	"	"	"
			162	"									"									"	"	A3 to B3	"	"	"
			163	"									"									"	"	A4 to B4	"	"	"
			164	"									"									"	"	A5 to B5	"	"	"
			165	"									"									"	"	A6 to B6	"	"	"
			166	"									"									"	"	A7 to B7	"	"	"
			167	"									"									"	"	A8 to B8	"	"	"
			168	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	IN	"	"	B1 to A1	"	"	"
			169	"									"									"	"	B2 to A2	"	"	"
			170	"									"									"	"	B3 to A3	"	"	"
			171	"									"									"	"	B4 to A4	"	"	"
			172	"									"									"	"	B5 to A5	"	"	"
			173	"									"									"	"	B6 to A6	"	"	"
			174	"									"									"	"	B7 to A7	"	"	"
			175	"									"									"	"	B8 to A8	"	"	"
	t_{PZH}	3003 (fig. 5)	176	4.5 V	GND	GND	GND	GND	GND	GND	GND	GND	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	IN	"	$\overline{G}$ to B1	6	25	"
			177	"									"									"	"	$\overline{G}$ to B2	"	"	"
			178	"									"									"	"	$\overline{G}$ to B3	"	"	"
			179	"									"									"	"	$\overline{G}$ to B4	"	"	"
			180	"									"									"	"	$\overline{G}$ to B5	"	"	"
			181	"									"									"	"	$\overline{G}$ to B6	"	"	"
			182	"									"									"	"	$\overline{G}$ to B7	"	"	"
			183	"									"									"	"	$\overline{G}$ to B8	"	"	"

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,X,2 Test no.	1 DIR	2 A1	3 A2	4 A3	5 A4	6 A5	7 A6	8 A7	9 A8	10 GND	11 B8	12 B7	13 B6	14 B5	15 B4	16 B3	17 B2	18 B1	19 $\overline{G}$	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
9 T_C = +25°C	t_{PZH}	3003 (fig. 5)	184	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	IN	5.0 V	G to A1	6	25	ns
			185	"									"											G to A2	"	"	"
			186	"									"											G to A3	"	"	"
			187	"									"											G to A4	"	"	"
			188	"									"											G to A5	"	"	"
			189	"									"											G to A6	"	"	"
			190	"									"											G to A7	"	"	"
			191	"									"											G to A8	"	"	"
	t_{PZL}	3003 (fig. 5)	192	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	"	G to B1	8	28	"
			193	"									"											G to B2	"	"	"
			194	"									"											G to B3	"	"	"
			195	"									"											G to B4	"	"	"
			196	"									"											G to B5	"	"	"
			197	"									"											G to B6	"	"	"
			198	"									"											G to B7	"	"	"
			199	"									"											G to B8	"	"	"
			200	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	"	"	G to A1	"	"	"
			201	"									"											G to A2	"	"	"
			202	"									"											G to A3	"	"	"
			203	"									"											G to A4	"	"	"
			204	"									"											G to A5	"	"	"
			205	"									"											G to A6	"	"	"
			206	"									"											G to A7	"	"	"
			207	"									"											G to A8	"	"	"
	t_{PHZ}	3003 (fig. 5)	208	4.5 V	GND	GND	GND	GND	GND	GND	GND	GND	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	"	G to B1	2	14	"
			209	"									"											G to B2	"	"	"
			210	"									"											G to B3	"	"	"
			211	"									"											G to B4	"	"	"
			212	"									"											G to B5	"	"	"
			213	"									"											G to B6	"	"	"
			214	"									"											G to B7	"	"	"
			215	"									"											G to B8	"	"	"
			216	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	4.5 V	"	"	G to A1	"	"	"
			217	"									"											G to A2	"	"	"
			218	"									"											G to A3	"	"	"
			219	"									"											G to A4	"	"	"
			220	"									"											G to A5	"	"	"
			221	"									"											G to A6	"	"	"
			222	"									"											G to A7	"	"	"
			223	"									"											G to A8	"	"	"
	t_{PLZ}	3003 (fig. 5)	224	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	"	G to B1	4	30	"
			225	"									"											G to B2	"	"	"
			226	"									"											G to B3	"	"	"
			227	"									"											G to B4	"	"	"
			228	"									"											G to B5	"	"	"
			229	"									"											G to B6	"	"	"
			230	"									"											G to B7	"	"	"
			231	"									"											G to B8	"	"	"
			232	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	"	"	G to A1	"	"	"
			233	"									"											G to A2	"	"	"
			234	"									"											G to A3	"	"	"
			235	"									"											G to A4	"	"	"
			236	"									"											G to A5	"	"	"
			237	"									"											G to A6	"	"	"
			238	"									"											G to A7	"	"	"
			239	"									"											G to A8	"	"	"

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,X,2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits		Unit
			Test no.	DIR	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	$\overline{G}$	V_{CC}		Min	Max	
10 T_C = +125°C	t_{PLH1}	3003 (fig. 5)	Same tests and terminal conditions as subgroup 9, except T_C = +125°C.																						3	15	"
	t_{PHL1}																								2	13	"
	t_{PZH}																								6	28	"
	t_{PZL}																								8	28	"
	t_{PHZ}																								2	17	"
	t_{PLZ}																								4	33	"
11	Same tests, terminal conditions, and limits as subgroup 10 except T_C = -55°C																										

1/ MIL-STD-883, method 3011 shall be used, except the output voltage shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one-half of the true circuit current I_{OS}.

2/ I_O limits for circuit C shall be -30 mA to -110 mA.

3/ A = 3.0 V minimum, B = 0.0 V or GND.

4/ H > 1.5 V; L < 1.5 V.

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TABLE III. Group A inspection for device type 05.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,X,Z	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits		Unit
			Test no.	DIR	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	$\overline{G}$	V_{CC}		Min	Max	
1 T_C = +25°C	V_{OH1}	3006	1	2.0 V	2.0 V								GND									0.8 V	4.5 V	B1	2.4		V
			2	"		2.0 V							"							-3 mA	-3 mA	"	"	B2	"		"
			3	"			2.0 V						"						-3 mA			"	"	B3	"		"
			4	"				2.0 V					"					-3 mA				"	"	B4	"		"
			5	"					2.0 V				"				-3 mA					"	"	B5	"		"
			6	"						2.0 V			"			-3 mA						"	"	B6	"		"
			7	"							2.0 V		"		-3 mA							"	"	B7	"		"
			8	"								2.0 V	"	-3 mA								"	"	B8	"		"
			9	0.8 V	-3 mA								"									"	"	A1	"		"
			10	"		-3 mA							"							2.0 V	2.0 V	"	"	A2	"		"
			11	"			-3 mA						"						2.0 V			"	"	A3	"		"
			12	"				-3 mA					"					2.0 V				"	"	A4	"		"
			13	"					-3 mA				"				2.0 V					"	"	A5	"		"
			14	"						-3 mA			"			2.0 V						"	"	A6	"		"
			15	"							-3 mA		"		2.0 V							"	"	A7	"		"
			16	"								-3 mA	"	2.0 V								"	"	A8	"		"
	V_{OH2}	3006	17	2.0 V	2.0 V								"							-12 mA	-12 mA	"	"	B1	2.0		"
			18	"		2.0 V							"						-12 mA			"	"	B2	"		"
			19	"			2.0 V						"					-12 mA				"	"	B3	"		"
			20	"				2.0 V					"				-12 mA					"	"	B4	"		"
			21	"					2.0 V				"			-12 mA						"	"	B5	"		"
			22	"						2.0 V			"		-12 mA							"	"	B6	"		"
			23	"							2.0 V		"	-12 mA								"	"	B7	"		"
			24	"								2.0 V	"									"	"	B8	"		"
			25	0.8 V	-12 mA								"							2.0 V	2.0 V	"	"	A1	"		"
			26	"		-12 mA							"						2.0 V			"	"	A2	"		"
			27	"			-12 mA						"					2.0 V				"	"	A3	"		"
			28	"				-12 mA					"				2.0 V					"	"	A4	"		"
			29	"					-12 mA				"			2.0 V						"	"	A5	"		"
			30	"						-12 mA			"		2.0 V							"	"	A6	"		"
			31	"							-12 mA		"	2.0 V								"	"	A7	"		"
			32	"								-12 mA	"		2.0 V							"	"	A8	"		"
	V_{OL}	3007	33	2.0 V	0.8 V								"							12 mA	12 mA	"	"	B1		0.4	"
			34	"		0.8 V							"						12 mA			"	"	B2		"	"
			35	"			0.8 V						"					12 mA				"	"	B3		"	"
			36	"				0.8 V					"				12 mA					"	"	B4		"	"
			37	"					0.8 V				"			12 mA						"	"	B5		"	"
			38	"						0.8 V			"		12 mA							"	"	B6		"	"
			39	"							0.8 V		"	12 mA								"	"	B7		"	"
			40	"								0.8 V	"									"	"	B8		"	"
			41	0.8 V	12 mA								"							0.8 V	0.8 V	"	"	A1		"	"
			42	"		12 mA							"						0.8 V			"	"	A2		"	"
			43	"			12 mA						"					0.8 V				"	"	A3		"	"
			44	"				12 mA					"				0.8 V					"	"	A4		"	"
			45	"					12 mA				"			0.8 V						"	"	A5		"	"
			46	"						12 mA			"		0.8 V							"	"	A6		"	"
			47	"							12 mA		"	0.8 V								"	"	A7		"	"
			48	"								12 mA	"		0.8 V							"	"	A8		"	"
	V_{IC}		49	-18 mA	-18 mA								"									"	"	DIR		-1.5	"
			50	"		-18 mA							"									"	"	A1		"	"
			51	"			-18 mA						"									"	"	A2		"	"
			52	"				-18 mA					"									"	"	A3		"	"
			53	"					-18 mA				"									"	"	A4		"	"
			54	"						-18 mA			"									"	"	A5		"	"
			55	"							-18 mA		"									"	"	A6		"	"
			56	"								-18 mA	"									"	"	A7		"	"
			57	"									"	-18 mA								"	"	A8		"	"
			58	"									"		-18 mA							"	"	B8		"	"
			59	"									"			-18 mA						"	"	B7		"	"
			60	"									"				-18 mA					"	"	B6		"	"
			61	"									"					-18 mA				"	"	B5		"	"
			62	"									"						-18 mA			"	"	B4		"	"
			63	"									"							-18 mA		"	"	B3		"	"
			64	"									"								-18 mA	"	"	B2		"	"
			65	"									"									"	"	B1		"	"
			66	"									"									"	"	$\overline{G}$		"	"

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,X,2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				DIR	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	$\overline{G}$	V_{CC}				
1 T_C = +25°C	I_{IL}	3009	67	0.4 V									GND										5.5 V	DIR	0	-100	µA
	I_{IL}	3009	68										"									0.4 V	"	$\overline{G}$	"	"	"
	I_{IL}/ I_{OZL}		69		0.4 V								"									4.5 V	"	A1	"	"	"
			70			0.4 V							"									"	"	A2	"	"	"
			71				0.4 V						"									"	"	A3	"	"	"
			72					0.4 V					"									"	"	A4	"	"	"
			73						0.4 V				"									"	"	A5	"	"	"
			74							0.4 V			"									"	"	A6	"	"	"
			75								0.4 V		"									"	"	A7	"	"	"
			76									0.4 V	"									"	"	A8	"	"	"
			77										"	0.4 V								"	"	B8	"	"	"
			78										"		0.4 V							"	"	B7	"	"	"
			79										"			0.4 V						"	"	B6	"	"	"
			80										"				0.4 V					"	"	B5	"	"	"
			81										"					0.4 V				"	"	B4	"	"	"
			82										"						0.4 V			"	"	B3	"	"	"
			83										"							0.4 V		"	"	B2	"	"	"
			84										"								0.4 V	"	"	B1	"	"	"
	I_{IH1}	3010	85	2.7 V									"									2.7 V	"	DIR		20	"
	I_{IH1}	3010	86										"										"	$\overline{G}$		"	"
	I_{IH1}/ I_{OZH}		87		2.7 V								"									4.5 V	"	A1		"	"
			88			2.7 V							"									"	"	A2		"	"
			89				2.7 V						"									"	"	A3		"	"
			90					2.7 V					"									"	"	A4		"	"
			91						2.7 V				"									"	"	A5		"	"
			92							2.7 V			"									"	"	A6		"	"
			93								2.7 V		"									"	"	A7		"	"
			94									2.7 V	"									"	"	A8		"	"
			95										"	2.7 V								"	"	B8		"	"
			96										"		2.7 V							"	"	B7		"	"
			97										"			2.7 V						"	"	B6		"	"
			98										"				2.7 V					"	"	B5		"	"
			99										"					2.7 V				"	"	B4		"	"
			100										"						2.7 V			"	"	B3		"	"
			101										"							2.7 V		"	"	B2		"	"
			102										"								2.7 V	"	"	B1		"	"
	I_{IH2}	3010	103		5.5 V								"										"	A1		100	"
			104			5.5 V							"										"	A2		"	"
			105				5.5 V						"										"	A3		"	"
			106					5.5 V					"										"	A4		"	"
			107						5.5 V				"										"	A5		"	"
			108							5.5 V			"										"	A6		"	"
			109								5.5 V		"										"	A7		"	"
			110									5.5 V	"										"	A8		"	"
			111										"	5.5 V									"	B8		"	"
			112										"		5.5 V								"	B7		"	"
			113										"			5.5 V							"	B6		"	"
			114										"				5.5 V						"	B5		"	"
			115										"					5.5 V					"	B4		"	"
			116										"						5.5 V				"	B3		"	"
			117										"							5.5 V			"	B2		"	"
			118										"								5.5 V		"	B1		"	"
	I_{IH3}	3010	119	7.0 V									"											DIR			
		3010	120										"									7.0 V	"	$\overline{G}$		"	"
	I_{O} 1/	3011	121	4.5 V	4.5 V								"												2/ -30	2/ -112	mA
			122	"		4.5 V							"											B1	"	"	"
			123	"			4.5 V						"											B2	"	"	"
			124	"				4.5 V					"											B3	"	"	"
			125	"					4.5 V				"											B4	"	"	"
			126	"						4.5 V			"											B5	"	"	"
			127	"							4.5 V		"											B6	"	"	"
			128	"								4.5 V	"	2.25 V										B7	"	"	"
															2.25 V									B8			
																2.25 V											
																	2.25 V										
																		2.25 V									
																			2.25 V								
																				2.25 V							
																					2.25 V						

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,X,2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				DIR	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	$\overline{G}$	V_{CC}				
1 T_C = +25°C	I_O 1/	3011	129	GND	2.25 V								GND									GND	5.5 V	A1	2/ -30	2/ -112	mA
			130	"	"	2.25 V							"									"	"	A2	"	"	"
			131	"	"		2.25 V						"						4.5 V			"	"	A3	"	"	"
			132	"	"			2.25 V					"					4.5 V				"	"	A4	"	"	"
			133	"	"				2.25 V				"				4.5 V					"	"	A5	"	"	"
			134	"	"					2.25 V			"			4.5 V						"	"	A6	"	"	"
			135	"	"						2.25 V		"		4.5 V							"	"	A7	"	"	"
			136	"	"							2.2 5V	"	4.5 V						4.5 V		"	"	A8	"	"	"
	I_{CCH}	3005	137	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"									"	"	V_{CC}		68	"
	I_{CCL}	3005	138	5.5 V	GND	GND	GND	GND	GND	GND	GND	GND	"									"	"	V_{CC}		60	"
	I_{CCZ}	3005	139	5.5 V									"									5.5 V	"	V_{CC}		63	"
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C and omit V_{IC} tests.																										
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and omit V_{IC} tests.																										
7 T_C = +25°C	Truth table tests	3014	140	A 3/	A	A	A	A	A	A	A	A	GND	H	H	H	H	H	H	H	H	B	4.5 V	See 4/			
			141	A	B 3/	B	B	B	B	B	B	B	"	L	L	L	L	L	L	L	L	"	"				
			142	B	H	H	H	H	H	H	H	H	"	A	A	A	A	A	A	A	A	"	"				
			143	B	L	L	L	L	L	L	L	L	"	B	B	B	B	B	B	B	B	"	"				
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																										
9 T_C = +25°C	t_{PLH1}	3003 (fig. 5)	144	4.5 V	IN	IN	IN	IN	IN	IN	IN	IN	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	5.0 V	A1 to B1	3	10	ns
			145	"									"									"	"	A2 to B2	"	"	"
			146	"									"									"	"	A3 to B3	"	"	"
			147	"									"									"	"	A4 to B4	"	"	"
			148	"									"									"	"	A5 to B5	"	"	"
			149	"									"									"	"	A6 to B6	"	"	"
			150	"									"									"	"	A7 to B7	"	"	"
			151	"									"									"	"	A8 to B8	"	"	"
			152	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	IN	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	IN	"	"	B1 to A1	"	"	"
			153	"									"									"	"	B2 to A2	"	"	"
			154	"									"									"	"	B3 to A3	"	"	"
			155	"									"									"	"	B4 to A4	"	"	"
			156	"									"									"	"	B5 to A5	"	"	"
			157	"									"									"	"	B6 to A6	"	"	"
			158	"									"									"	"	B7 to A7	"	"	"
			159	"									"									"	"	B8 to A8	"	"	"
	t_{PHL1}	3003 (fig. 5)	160	4.5 V	IN	IN	IN	IN	IN	IN	IN	IN	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	"	A1 to B1	"	"	"
			161	"									"									"	"	A2 to B2	"	"	"
			162	"									"									"	"	A3 to B3	"	"	"
			163	"									"									"	"	A4 to B4	"	"	"
			164	"									"									"	"	A5 to B5	"	"	"
			165	"									"									"	"	A6 to B6	"	"	"
			166	"									"									"	"	A7 to B7	"	"	"
			167	"									"									"	"	A8 to B8	"	"	"
			168	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	IN	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	IN	"	"	B1 to A1	"	"	"
			169	"									"									"	"	B2 to A2	"	"	"
			170	"									"									"	"	B3 to A3	"	"	"
			171	"									"									"	"	B4 to A4	"	"	"
			172	"									"									"	"	B5 to A5	"	"	"
			173	"									"									"	"	B6 to A6	"	"	"
			174	"									"									"	"	B7 to A7	"	"	"
			175	"									"									"	"	B8 to A8	"	"	"
	t_{PZH}	3003 (fig. 5)	176	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	IN	"	$\overline{G}$ to B1	7	20	"
			177	"									"									"	"	$\overline{G}$ to B2	"	"	"
			178	"									"									"	"	$\overline{G}$ to B3	"	"	"
			179	"									"									"	"	$\overline{G}$ to B4	"	"	"
			180	"									"									"	"	$\overline{G}$ to B5	"	"	"
			181	"									"									"	"	$\overline{G}$ to B6	"	"	"
			182	"									"									"	"	$\overline{G}$ to B7	"	"	"
			183	"									"									"	"	$\overline{G}$ to B8	"	"	"

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,X,Z	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits		Unit
			Test no.	DIR	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	$\overline{G}$	V_{CC}		Min	Max	
9 T_C = +25°C	t_{PZH}	3003 (fig. 5)	184	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	IN	5.0 V	G to A1	7	20	ns
			185	"									"									"	"	G to A2	"	"	
			186	"									"									"	"	G to A3	"	"	
			187	"									"									"	"	G to A4	"	"	
			188	"									"									"	"	G to A5	"	"	
			189	"									"									"	"	G to A6	"	"	
			190	"									"									"	"	G to A7	"	"	
			191	"									"									"	"	G to A8	"	"	
	t_{PZL}	3003 (fig. 5)	192	4.5 V	GND	GND	GND	GND	GND	GND	GND	GND	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	"	G to B1	10	26	"
			193	"									"									"	"	G to B2	"	"	
			194	"									"									"	"	G to B3	"	"	
			195	"									"									"	"	G to B4	"	"	
			196	"									"									"	"	G to B5	"	"	
			197	"									"									"	"	G to B6	"	"	
			198	"									"									"	"	G to B7	"	"	
			199	"									"									"	"	G to B8	"	"	
			200	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	GND	GND	GND	GND	GND	GND	GND	GND	"	"	G to A1	"	"	
			201	"									"									"	"	G to A2	"	"	
			202	"									"									"	"	G to A3	"	"	
			203	"									"									"	"	G to A4	"	"	
			204	"									"									"	"	G to A5	"	"	
			205	"									"									"	"	G to A6	"	"	
			206	"									"									"	"	G to A7	"	"	
			207	"									"									"	"	G to A8	"	"	
	t_{PHZ}	3003 (fig. 5)	208	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	"	G to B1	3	16	"
			209	"									"									"	"	G to B2	"	"	
			210	"									"									"	"	G to B3	"	"	
			211	"									"									"	"	G to B4	"	"	
			212	"									"									"	"	G to B5	"	"	
			213	"									"									"	"	G to B6	"	"	
			214	"									"									"	"	G to B7	"	"	
			215	"									"									"	"	G to B8	"	"	
			216	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	"	"	G to A1	"	"	
			217	"									"									"	"	G to A2	"	"	
			218	"									"									"	"	G to A3	"	"	
			219	"									"									"	"	G to A4	"	"	
			220	"									"									"	"	G to A5	"	"	
			221	"									"									"	"	G to A6	"	"	
			222	"									"									"	"	G to A7	"	"	
			223	"									"									"	"	G to A8	"	"	
	t_{PLZ}	3003 (fig. 5)	224	4.5 V	GND	GND	GND	GND	GND	GND	GND	GND	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	"	G to B1	4	18	"
			225	"									"									"	"	G to B2	"	"	
			226	"									"									"	"	G to B3	"	"	
			227	"									"									"	"	G to B4	"	"	
			228	"									"									"	"	G to B5	"	"	
			229	"									"									"	"	G to B6	"	"	
			230	"									"									"	"	G to B7	"	"	
			231	"									"									"	"	G to B8	"	"	
			232	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	GND	GND	GND	GND	GND	GND	GND	GND	"	"	G to A1	"	"	
			233	"									"									"	"	G to A2	"	"	
			234	"									"									"	"	G to A3	"	"	
			235	"									"									"	"	G to A4	"	"	
			236	"									"									"	"	G to A5	"	"	
			237	"									"									"	"	G to A6	"	"	
			238	"									"									"	"	G to A7	"	"	
			239	"									"									"	"	G to A8	"	"	

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,X,2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits		Unit
			Test no.	DIR	A1	A2	A3	A4	A5	A6	A7	A8	GND	B8	B7	B6	B5	B4	B3	B2	B1	$\overline{G}$	V_{CC}		Min	Max	
10 T_C = +125°C	t_{PLH1}	3003 (fig. 5)	Same tests and terminal conditions as subgroup 9, except T_C = +125°C.																						3	15	ns
	t_{PHL1}																								3	13	"
	t_{PHZ}																								7	25	"
	t_{PLZ}																								10	30	"
	t_{PZH}																								3	18	"
	t_{PZL}																								4	20	"
11	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C																										

1/ MIL-STD-883, method 3011 shall be used, except the output voltage shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one-half of the true short circuit current I_{OS}.

2/ I_{OS} limits for circuit C shall be -30 mA to -110 mA.

3/ A > 3.0 V minimum, B < 0.0 V or GND.

4/ H > 1.5 V; L < 1.5 V.

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TABLE III. Group A inspection for device type 06.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 1/ X and Z / Cases C and D	2 / 1	3 / 2	4 / 3	6 / 4	8 / 5	9 / 6	10 / 7	12 / 8	13 / 9	14 / 10	16 / 11	18 / 12	19 / 13	20 / 14	Measured terminal	Limits Min	Limits Max	Unit
			Test no.	$\overline{GAB}$	NC	A1	A2	A3	A4	GND	B4	B3	B2	B1	NC	GBA	V_{CC}				
1 T_C = +25°C	V_{OH1}	3006	1	0.8 V		0.8 V				GND			-3 mA	-3 mA		0.8 V	4.5 V	B1	2.4		V
			2	"			0.8 V			"			"			"	"	B2	"		"
			3	"				0.8 V		"		-3 mA				"	"	B3	"		"
			4	"					0.8 V	"	-3 mA					"	"	B4	"		"
			5	2.0 V		-3 mA				"				0.8 V		2.0 V	"	A1	"		"
			6	"			-3 mA			"			0.8 V			"	"	A2	"		"
			7	"				-3 mA		"		0.8 V				"	"	A3	"		"
			8	"					-3 mA	"	0.8 V					"	"	A4	"		"
	V_{OH2}	3006	9	0.5 V		0.5 V				"			-12 mA	-12 mA		0.5 V	"	B1	2.0		"
			10	"			0.5 V			"			"			"	"	B2	"		"
			11	"				0.5 V		"		-12 mA				"	"	B3	"		"
			12	"					0.5 V	"	-12 mA					"	"	B4	"		"
			13	2.0 V		-12 mA				"			0.5 V	0.5 V		2.0 V	"	A1	"		"
			14	"			-12 mA			"			"			"	"	A2	"		"
			15	"				-12 mA		"		0.5 V				"	"	A3	"		"
			16	"					-12 mA	"	0.5 V					"	"	A4	"		"
	V_{OL}	3007	17	0.8 V		2.0 V				"			12 mA	12 mA		0.8 V	"	B1		0.4	"
			18	"			2.0 V			"			"			"	"	B2		"	"
			19	"				2.0 V		"		12 mA				"	"	B3		"	"
			20	"					2.0 V	"	12 mA					"	"	B4		"	"
			21	2.0 V		12 mA				"			2.0 V	2.0 V		2.0 V	"	A1		"	"
			22	"			12 mA			"			"			"	"	A2		"	"
			23	"				12 mA		"		2.0 V				"	"	A3		"	"
			24	"					12 mA	"	2.0 V					"	"	A4		"	"
	V_{IC}		25	-18 mA		-18 mA				"							"	$\overline{GAB}$		-1.5	"
			26				-18 mA			"							"	A1		"	"
			27					-18 mA		"							"	A2		"	"
			28						-18 mA	"							"	A3		"	"
			29							"	-18 mA						"	A4		"	"
			30							"		-18 mA					"	B4		"	"
			31							"			-18 mA				"	B3		"	"
			32							"				-18 mA			"	B2		"	"
			33							"							"	B1		"	"
			34							"						-18 mA	"	GBA		"	"
	I_{IL}	3009	35	0.4 V						"							5.5 V	$\overline{GAB}$	0	-100	uA
	I_{IL}	3009	36	"						"						0.4 V	"	GBA	"	"	"
	I_{OZL}/ I_{IL}	3009	37	2.0 V		0.4 V				"						0.8 V	"	A1	"	"	"
			38	"			0.4 V			"						"	"	A2	"	"	"
			39	"				0.4 V		"						"	"	A3	"	"	"
			40	"					0.4 V	"						"	"	A4	"	"	"
			41	"						"	0.4 V					"	"	B4	"	"	"
			42	"						"		0.4 V				"	"	B3	"	"	"
			43	"						"			0.4 V			"	"	B2	"	"	"
			44	"						"				0.4 V		"	"	B1	"	"	"
	I_{IH1}	3010	45	2.7 V						"							"	GAB		20	"
	I_{IH1}	3010	46	"						"						2.7 V	"	GBA		"	"
	I_{OZH}/ I_{IH1}	3010	47	2.0 V		2.7 V				"						0.8 V	"	A1		"	"
			48	"			2.7 V			"						"	"	A2		"	"
			49	"				2.7 V		"						"	"	A3		"	"
			50	"					2.7 V	"						"	"	A4		"	"
			51	"						"	2.7 V					"	"	B4		"	"
			52	"						"		2.7 V				"	"	B3		"	"
			53	"						"			2.7 V			"	"	B2		"	"
			54	"						"				2.7 V		"	"	B1		"	"

See footnotes at end of device type 06.

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 1/ X and Z / Cases C and D / Test no.	2 / 1 / GAB	3 / 2 / NC	4 / 3 / A1	6 / 4 / A2	8 / 5 / A3	9 / 6 / A4	10 / 7 / GND	12 / 8 / B4	13 / 9 / B3	14 / 10 / B2	16 / 11 / B1	18 / 12 / NC	19 / 13 / GBA	20 / 14 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	I_{IH2}	3010	55			5.5 V				GND							5.5 V	A1		100	µA
			56				5.5 V			"							"	A2		"	"
			57					5.5 V		"							"	A3		"	"
			58						5.5 V	"							"	A4		"	"
			59							"		5.5 V					"	B4		"	"
			60							"			5.5 V				"	B3		"	"
			61							"				5.5 V			"	B2		"	"
			62							"					5.5 V		"	B1		"	"
	I_{IH3}	3010	63	7.0 V						"							"	$\overline{GAB}$		"	"
	I_{IH3}	3010	64							"						7.0 V	"	GBA		"	"
	I_O 2/	3011	65	GND		GND				"				2.25 V			"	B1	3/ -30	3/ -112	mA
			66				GND			"			2.25 V			GND	"	B2	"	"	"
			67					GND		"		2.25 V					"	B3	"	"	"
			68						GND	"	2.25 V						"	B4	"	"	"
			69	5.5 V		2.25 V				"				GND		5.5 V	"	A1	"	"	"
			70				2.25 V			"			GND				"	A2	"	"	"
			71					2.25 V		"		GND					"	A3	"	"	"
			72						2.25 V	"	GND						"	A4	"	"	"
	I_{CCH}	3005	73	GND		GND	GND	GND	GND	"						GND	"	V_{CC}		20	"
	I_{CCL}	3005	74	GND		4.5 V	4.5 V	4.5 V	4.5 V	"						GND	"	V_{CC}		26	"
	I_{CCZ}	3005	75	5.5 V						"						GND	"	V_{CC}		27	"
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C and omit V_{IC} tests.																				
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and omit V_{IC} tests.																				
7 T_C = +25°C	Truth table tests	3014 4/	76	B		A	A	A	A	GND	L	L	L	L		B	5.0 V	See 5/			
			77	B		A	B	B	B	"	H	H	H	H		B	"				
			78	A		L	L	L	L	"	A	A	A	A		A	"				
			79	A		H	H	H	H	"	B	B	B	B		B	"				
8	Same tests, and terminal conditions as subgroup 7, except T_C = +125°C and -55°C.																				
9 T_C = +25°C	t_{PLH1}	3003 (fig. 5)	80	GND		IN				GND				OUT		GND	5.0 V	A1 to B1	3	11	ns
			81	"			IN			"			OUT			"	"	A2 to B2	"	"	"
			82	"				IN		"		OUT				"	"	A3 to B3	"	"	"
			83	"					IN	"	OUT					"	"	A4 to B4	"	"	"
			84	4.5 V		OUT				"				IN		4.5 V	"	B1 to A1	"	"	"
			85	"			OUT			"			IN			"	"	B2 to A2	"	"	"
			86	"				OUT		"		IN				"	"	B3 to A3	"	"	"
			87	"					OUT	"	IN					"	"	B4 to A4	"	"	"
	t_{PHL1}	3003 (fig. 5)	88	GND		IN				"				OUT		GND	"	A1 to B1	2	10	"
			89	"			IN			"			OUT			"	"	A2 to B2	"	"	"
			90	"				IN		"		OUT				"	"	A3 to B3	"	"	"
			91	"					IN	"	OUT					"	"	A4 to B4	"	"	"
			92	4.5 V		OUT				"				IN		4.5 V	"	B1 to A1	"	"	"
			93	"			OUT			"			IN			"	"	B2 to A2	"	"	"
			94	"				OUT		"		IN				"	"	B3 to A3	"	"	"
			95	"					OUT	"	IN					"	"	B4 to A4	"	"	"
	t_{PZH}	3003 (fig. 5)	96	IN		GND				"				OUT		GND	"	$\overline{GAB}$ to B1	4	18	"
			97	"			GND			"			OUT			"	"	$\overline{GAB}$ to B2	"	"	"
			98	"				GND		"		OUT				"	"	$\overline{GAB}$ to B3	"	"	"
			99	"					GND	"	OUT					"	"	$\overline{GAB}$ to B4	"	"	"
			100	4.5 V		OUT				"				GND		IN	"	GBA to A1	"	"	"
			101	"			OUT			"			GND			"	"	GBA to A2	"	"	"
			102	"				OUT		"		GND				"	"	GBA to A3	"	"	"
			103	"					OUT	"	GND					"	"	GBA to A4	"	"	"

See footnotes at end of device type 06.

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 1/ X and Z / Cases C and D	2	3	4	6	8	9	10	12	13	14	16	18	19	20	Measured terminal	Limits		Unit
			Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14		Min	Max	
				$\overline{GAB}$	NC	A1	A2	A3	A4	GND	B4	B3	B2	B1	NC	GBA	V_{CC}				
9 T_C = +25°C	t_{PZL}	3003 (fig. 5)	104	IN		4.5 V				GND				OUT		GND	5.0 V	$\overline{GAB}$ to B1	7	21	ns
			105	"			4.5 V			"			OUT			"	"	$\overline{GAB}$ to B2	"	"	"
			106	"				4.5 V		"		OUT				"	"	$\overline{GAB}$ to B3	"	"	"
			107	"					4.5 V	"	OUT					"	"	$\overline{GAB}$ to B4	"	"	"
			108	4.5 V		OUT				"				4.5 V		IN	"	GBA to A1	"	"	"
			109	"			OUT			"			4.5 V			"	"	GBA to A2	"	"	"
			110	"				OUT		"		4.5 V				"	"	GBA to A3	"	"	"
			111	"					OUT	"	4.5 V					"	"	GBA to A4	"	"	"
	t_{PHZ}	3003 (fig. 5)	112	IN		GND				"				OUT		GND	"	$\overline{GAB}$ to B1	2	14	"
			113	"			GND			"			OUT			"	"	$\overline{GAB}$ to B2	"	"	"
			114	"				GND		"		OUT				"	"	$\overline{GAB}$ to B3	"	"	"
			115	"					GND	"	OUT					"	"	$\overline{GAB}$ to B4	"	"	"
			116	4.5 V		OUT				"				GND		IN	"	GBA to A1	"	"	"
			117	"			OUT			"			GND			"	"	GBA to A2	"	"	"
			118	"				OUT		"		GND				"	"	GBA to A3	"	"	"
			119	"					OUT	"	GND					"	"	GBA to A4	"	"	"
	t_{PLZ}	3003 (fig. 5)	120	IN		4.5 V				"				OUT		GND	"	$\overline{GAB}$ to B1	4	22	"
			121	"			4.5 V			"			OUT			"	"	$\overline{GAB}$ to B2	"	"	"
			122	"				4.5 V		"		OUT				"	"	$\overline{GAB}$ to B3	"	"	"
			123	"					4.5 V	"	OUT					"	"	$\overline{GAB}$ to B4	"	"	"
			124	4.5 V		OUT				"				4.5 V		IN	"	GBA to A1	"	"	"
			125	"			OUT			"			4.5 V			"	"	GBA to A2	"	"	"
			126	"				OUT		"		4.5 V				"	"	GBA to A3	"	"	"
			127	"					OUT	"	4.5 V					"	"	GBA to A4	"	"	"
10 T_C = +125°C	t_{PLH}	Same tests and terminal conditions as subgroup 9; except T_C = +125°C.																	3	15	"
	t_{PHL}																		2	14	"
	t_{PZH}																		4	22	"
	t_{PZL}																		7	25	"
	t_{PHZ}																		2	16	"
	t_{PLZ}																		4	28	"
11 T_C = -55°C	Same tests, terminal conditions and limits as subgroup 10, except T_C = -55°C.																				

1/ Pins not referenced are NC.

2/ MIL-STD-883, method 3011 shall be used, except the output voltage shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one-half of the true short circuit current I_{OS}.

3/ I_O limits for circuit C shall be -30 mA to -110 mA.

4/ A = 3.0 V minimum, B = 0.0 V or GND.

5/ H > 1.5 V; L < 1.5 V.

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TABLE III. Group A inspection for device type 07.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 1/ X and Z / Cases C and D	2	3	4	6	8	9	10	12	13	14	16	18	19	20	Measured terminal	Limits		Unit
				1	2	3	4	5	6	7	8	9	10	11	12	13	14		Min	Max	
			Test no.	GAB	NC	A1	A2	A3	A4	GND	B4	B3	B2	B1	NC	GBA	V_{CC}				
1 T_C = +25°C	V_{OH1}	3006	1	0.8 V		2.0 V				GND				-3 mA		0.8 V	4.5 V	B1	2.4		V
			2				2.0 V			"			-3 mA				"	B2	"		"
			3					2.0 V		"		-3 mA					"	B3	"		"
			4						2.0 V	"	-3 mA						"	B4	"		"
			5	2.0 V		-3 mA				"				2.0 V		2.0 V	"	A1	"		"
			6				-3 mA			"			-3 mA				"	A2	"		"
			7					-3 mA		"		2.0 V					"	A3	"		"
			8						-3 mA	"	2.0 V						"	A4	"		"
	V_{OH2}	3006	9	0.5 V		2.0 V				"				-12 mA		0.5 V	"	B1	2.0		"
			10				2.0 V			"			-12 mA				"	B2	"		"
			11					2.0 V		"		-12 mA					"	B3	"		"
			12						2.0 V	"	-12 mA						"	B4	"		"
			13	2.0 V		-12 mA				"				2.0 V		2.0 V	"	A1	"		"
			14				-12 mA			"			-12 mA				"	A2	"		"
			15					-12 mA		"		2.0 V					"	A3	"		"
			16						-12 mA	"	2.0 V						"	A4	"		"
	V_{OL}	3007	17	0.8 V		0.8 V				"				12 mA		0.8 V	"	A1		0.4	"
			18	"			0.8 V			"			12 mA			"	"	A2		"	"
			19	"				0.8 V		"		12 mA					"	A3		"	"
			20						0.8 V	"	12 mA						"	A4		"	"
			21	2.0 V		12 mA				"				0.8 V		2.0 V	"	B1		"	"
			22	"			12 mA			"			0.8 V			"	"	B2		"	"
			23	"				12 mA		"		0.8 V					"	B3		"	"
			24	"					12 mA	"	0.8 V						"	B4		"	"
	V_{IC}		25	-18 mA		-18 mA				"							"	GAB		-1.5	"
			26				-18 mA			"							"	A1		"	"
			27					-18 mA		"							"	A2		"	"
			28						-18 mA	"	-18 mA						"	A3		"	"
			29							"		-18 mA					"	A4		"	"
			30							"			-18 mA				"	B4		"	"
			31							"				-18 mA			"	B3		"	"
			32							"							"	B2		"	"
			33							"							"	B1		"	"
			34							"						-18 mA	"	GBA		"	"
	I_{IL}	3009	35	0.4 V						"							5.5 V	GAB	0	-100	µA
	I_{IL}	3009	36							"						0.4 V	"	GBA	"	"	"
	I_{IL}/ I_{OZL}		37	4.5 V		0.4 V				"						GND	"	A1	"	"	"
			38	"			0.4 V			"						"	"	A2	"	"	"
			39	"				0.4 V		"						"	"	A3	"	"	"
			40	"					0.4 V	"						"	"	A4	"	"	"
			41	"						"	0.4 V					"	"	B4	"	"	"
			42	"						"		0.4 V				"	"	B3	"	"	"
			43	"						"			0.4 V			"	"	B2	"	"	"
			44	"						"				0.4 V		"	"	B1	"	"	"
	I_{IH1}	3010	45	2.7 V						"							"	GAB		20	"
	I_{IH1}	3010	46	"						"						2.7 V	"	GBA		"	"
	I_{IH1}/ I_{OZH}		47	4.5 V		2.7 V				"						4.5 V	"	A1		"	"
			48	"			2.7 V			"						"	"	A2		"	"
			49	"				2.7 V		"						"	"	A3		"	"
			50	"					2.7 V	"						"	"	A4		"	"
			51	"						"	2.7 V					"	"	B4		"	"
			52	"						"		2.7 V				"	"	B3		"	"
			53	"						"			2.7 V			"	"	B2		"	"
			54	"						"				2.7 V		"	"	B1		"	"

See footnotes at end of device type 07.

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TABLE III. Group A inspection for device type 07 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 1/ X and Z	2	3	4	6	8	9	10	12	13	14	16	18	19	20	Measured terminal	Limits		Unit
			Cases C and D	1	2	3	4	5	6	7	8	9	10	11	12	13	° 14		Min	Max	
			Test no.	GAB	NC	A1	A2	A3	A4	GND	B4	B3	B2	B1	NC	GBA	V_{CC}				
1 T_C = +25°C	I_{IH2}	3010	55			5.5 V	5.5 V	5.5 V	5.5 V	GND		5.5 V	5.5 V	5.5 V	5.5 V		5.5 V	A1		100	µA
			56							"							"	A2		"	"
			57							"							"	A3		"	"
			58							"							"	A4		"	"
			59							"							"	B4		"	"
			60							"							"	B3		"	"
			61							"							"	B2		"	"
			62							"							"	B1		"	"
	I_{IH3}	3010	63	7.0 V						"						7.0 V	"	GAB		"	"
	I_{IH3}	3010	64							"							"	GBA		"	"
	I_O 2/	3011	65	GND		5.5 V	5.5 V	5.5 V	5.5 V	"	2.25 V	2.25 V	2.25 V	2.25 V		GND	"	B1	3/ -30	3/ -112	mA
			66	"						"						"	"	B2	"	"	"
			67	"						"						"	"	B3	"	"	"
			68	"						"						"	"	B4	"	"	"
			69	5.5 V		2.25 V	2.25 V	2.25 V	2.25 V	"	5.5 V	5.5 V	5.5 V	5.5 V		5.5 V	"	A1	"	"	"
			70	"						"						"	"	A2	"	"	"
			71	"						"						"	"	A3	"	"	"
			72	"						"						"	"	A4	"	"	"
	I_{CCH}	3005	73	GND		4.5 V	4.5 V	4.5 V	4.5 V	"						GND	"	V_{CC}		25	"
	I_{CCL}	3005	74	GND		GND	GND	GND	GND	"						GND	"	V_{CC}		30	"
	I_{CCZ}	3005	75	4.5 V		4.5 V	4.5 V	4.5 V	4.5 V	"						4.5 V	"	V_{CC}		32	"
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C and omit V_{IC} tests.																				
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and omit V_{IC} tests.																				
7 T_C = +25°C	Truth table tests	3014 4/	76	B		A	A	A	A	GND	H	H	H	H		B	4.5 V	See 5/			
			77	B		H	H	H	H	"	L	L	L	L		A	"				
			78	A		H	H	H	H	"	A	A	A	A		A	"				
			79	A		L	L	L	L	"	B	B	B	B		A	"				
8	Same tests and terminal conditions as subgroup 7, except T_C = +125°C and -55°C.																				
9 T_C = +25°C	t_{PLH1}	3003 (fig. 5)	80	GND		IN	IN	IN	IN	GND	OUT	OUT	OUT	OUT		GND	5.0 V	A1 to B1	4	11	ns
			81	"						"						"	"	A2 to B2	"	"	"
			82	"						"						"	"	A3 to B3	"	"	"
			83	"						"						"	"	A4 to B4	"	"	"
			84	"		OUT	OUT	OUT	OUT	"	IN	IN	IN	IN		4.5 V	"	B1 to A1	"	"	"
			85	"						"						"	"	B2 to A2	"	"	"
			86	"						"						"	"	B3 to A3	"	"	"
			87	4.5 V						"						"	"	B4 to A4	"	"	"
	t_{PHL1}	3003 (fig. 5)	88	GND		IN	IN	IN	IN	"	OUT	OUT	OUT	OUT		GND	"	A1 to B1	"	"	"
			89	"						"						"	"	A2 to B2	"	"	"
			90	"						"						"	"	A3 to B3	"	"	"
			91	"						"						"	"	A4 to B4	"	"	"
			92	"		OUT	OUT	OUT	OUT	"	IN	IN	IN	IN		4.5 V	"	B1 to A1	"	"	"
			93	"						"						"	"	B2 to A2	"	"	"
			94	"						"						"	"	B3 to A3	"	"	"
			95	4.5 V						"						"	"	B4 to A4	"	"	"
	t_{PZH}	3003 (fig. 5)	96	IN		4.5 V	4.5 V	4.5 V	4.5 V	"	OUT	OUT	OUT	OUT		GND	"	GAB to B1	7	20	"
			97	"						"						"	"	GAB to B2	"	"	"
			98	"						"						"	"	GAB to B3	"	"	"
			99	"						"						"	"	GAB to B4	"	"	"
			100	"		OUT	OUT	OUT	OUT	"	4.5 V	4.5 V	4.5 V	4.5 V		IN	"	GBA to A1	"	"	"
			101	"						"						"	"	GBA to A2	"	"	"
			102	"						"						"	"	GBA to A3	"	"	"
			103	4.5 V						"						"	"	GBA to A4	"	"	"

See footnotes at end of device type 07.

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TABLE III. Group A inspection for device type 07 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 1/ X and Z / Cases C and D	2	3	4	6	8	9	10	12	13	14	16	18	19	20	Measured terminal	Limits		Unit
				1	2	3	4	5	6	7	8	9	10	11	12	13	14				
			Test no.	GAB	NC	A1	A2	A3	A4	GND	B4	B3	B2	B1	NC	GBA	V_{CC}		Min	Max	
9 T_C = +25°C	t_{PZL}	3003 (fig. 5)	104	IN		GND	GND	GND	GND	GND	OUT	OUT	OUT	OUT		GND	5.0 V	GAB to B1	7	20	ns
			105							"						"	"	GAB to B2	"	"	"
			106							"						"	"	GAB to B3	"	"	"
			107							"						"	"	GAB to B4	"	"	"
			108	4.5 V		OUT			GND	"	OUT			GND		IN	"	GBA to A1	"	"	"
			109				OUT			"			GND			"	"	GBA to A2	"	"	"
			110					OUT		"		GND				"	"	GBA to A3	"	"	"
			111						OUT	"	GND					"	"	GBA to A4	"	"	"
	t_{PHZ}	3003 (fig. 5)	112	IN		4.5 V	4.5 V	4.5 V	4.5 V	"			OUT	OUT		GND	"	GAB to B1	2	14	"
			113							"						"	"	GAB to B2	"	"	"
			114							"						"	"	GAB to B3	"	"	"
			115							"						"	"	GAB to B4	"	"	"
			116	4.5 V		OUT			4.5 V	"	OUT			4.5 V		IN	"	GBA to A1	"	"	"
			117				OUT			"		OUT				"	"	GBA to A2	"	"	"
			118					OUT		"			4.5 V			"	"	GBA to A3	"	"	"
			119						OUT	"	4.5 V	4.5 V				"	"	GBA to A4	"	"	"
	t_{PLZ}	3003 (fig. 5)	120	IN		GND	GND	GND	GND	"			OUT	OUT		GND	"	GAB to B1	4	25	"
			121							"						"	"	GAB to B2	"	"	"
			122							"						"	"	GAB to B3	"	"	"
			123							"						"	"	GAB to B4	"	"	"
			124	4.5 V		OUT			GND	"	OUT			GND		IN	"	GBA to A1	"	"	"
			125				OUT			"		OUT				"	"	GBA to A2	"	"	"
			126					OUT		"			GND			"	"	GBA to A3	"	"	"
			127						OUT	"	GND	GND				"	"	GBA to A4	"	"	"
10 T_C = +125°C	t_{PLH}	Same tests and terminal conditions as subgroup 9, except T_C = +125°C.																	4	15	"
	t_{PHL}																		4	15	"
	t_{PZH}																		7	25	"
	t_{PZL}																		7	25	"
	t_{PHZ}																		2	16	"
	t_{PLZ}																		4	30	"
11 T_C = -55°C	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																				

1/ Pins not referenced are NC.

2/ MIL-STD-883, method 3011 shall be used, except the output voltage shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one-half of the true short circuit current I_{OS}.

3/ I_O limits for circuit C shall be -30 mA to -110 mA.

4/ A = 3.0 V minimum, B = 0.0 V or GND.

5/ H > 1.5 V; L < 1.5 V.

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g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND	Ground zero voltage potential.
V_{IN}	Voltage level at an input terminal.
I_{IN}	Current flowing into an input terminal.
t_{PHZ}	Output disable time (of a three-state output) from high level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from the defined high level to a high impedance (off) state.
t_{PLZ}	Output disable time (of a three-state output) from low level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from the defined low level to a high-impedance (off) state.
t_{PZH}	Output enable time (of a three-state output) to high level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from a high impedance (off) state to the defined high level.
t_{PZL}	Output enable time (of a three-state output) to low level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from a high-impedance (off) state to the defined low level.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54ALS640
02	54ALS641
03	54ALS642
04	54ALS643
05	54ALS645
06	54ALS242
07	54ALS243

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6.6 Manufacturers' designators. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designations.

Device type	Circuits		
	A	B	C
	Texas Instruments	Motorola Incorporated	National Semiconductor Corporation
01	X		
02	X		
03	X		
04	X		
05	X		
06	X		
07	X		

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1106)

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